CEEDS_ValidatedHistoricalData Document

General

CEEDS_ValidatedHistoricalData Document contextual model

General

!# shows the diagram.

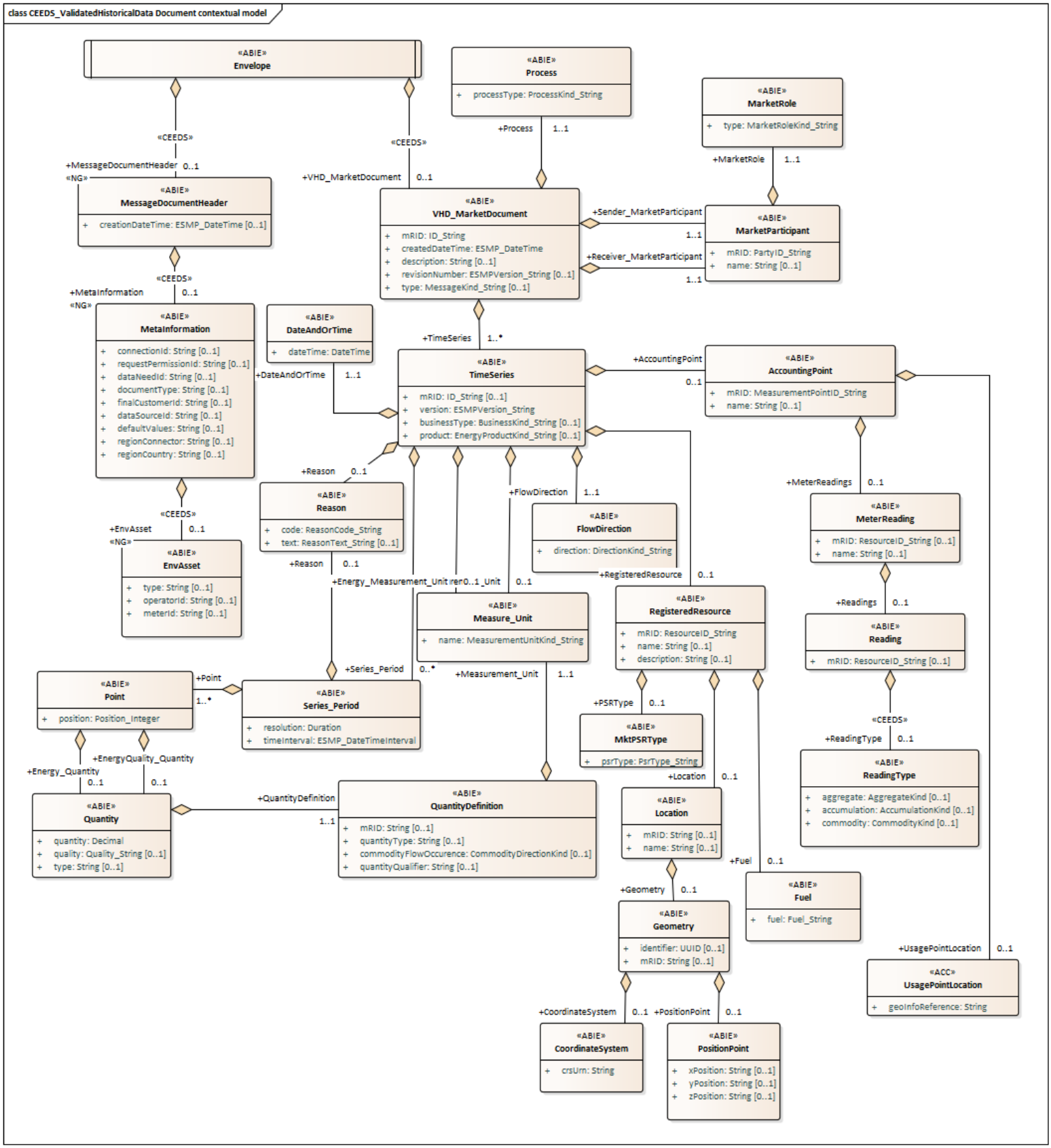


Figure !# - diagram CEEDS_ValidatedHistoricalData Document contextual model

(ABIE) AccountingPoint

An administrative point where Energy Supplier change can take place and for which commercial business processes are defined.

!# shows all attributes of AccountingPoint.

Table !# - Attributes of CEEDS_ValidatedHistoricalData Document contextual model::AccountingPoint

Attribute name	mult.	Attribute type	Description
mRID (ABIE)	1..1	MeasurementPointID_String	A unique identification of the measurement point. In the ESMP context, the "model authority" is defined as an authorized issuing office that provides an agreed identification coding scheme for market participant, domain, measurement point, resources (generator, lines, substations, etc.) identification. Master resource identifier issued by a model authority. The mRID is globally unique within an exchange context. Global uniqueness is easily achieved by using a UUID for the mRID. It is strongly recommended to do this. For CIMXML data files in RDF syntax, the mRID is mapped to rdf:ID or rdf:about attributes that identify CIM object elements.
name (ABIE)	0..1	String	The name is any free human readable and possibly non unique text naming the object.

!# shows all association ends of AccountingPoint with other classes.

Table !# - Association ends of CEEDS_ValidatedHistoricalData Document contextual model::AccountingPoint with other classes

mult.	mult. Class type name	Description
UsagePointLocation (ABIE)	[0..1] UsagePointLocation	Location of this usage point. Association Based On : ESMPClasses:: UsagePointLocation.UsagePointLocation[0..1] ----- ESMPClasses:: MarketEvaluationPoint.[]
MeterReadings (ABIE)	[0..1] MeterReading	All meter readings obtained from this usage point. Association Based On : ESMPClasses:: MeterReading.MeterReadings[0..*] ----- ESMPClasses:: MarketEvaluationPoint.[]

(ABIE) CoordinateSystem

Coordinate reference system.

!# shows all attributes of CoordinateSystem.

Table !# - Attributes of CEEDS_ValidatedHistoricalData Document contextual model::CoordinateSystem

Attribute name	mult.	Attribute type	Description
crsUrn (ABIE)	1..1	String	A Uniform Resource Name (URN) for the coordinate reference system (crs) used to define 'Location.PositionPoints'. An example would be the European Petroleum Survey Group (EPSG) code for a coordinate reference system, defined in URN under the Open Geospatial Consortium (OGC) namespace as: urn:ogc:def:uom:EPSG::XXXX, where XXXX is an EPSG code (a full list of codes can be found at the EPSG Registry web site http://www.epsg-registry.org/). To define the coordinate system as being WGS84 (latitude, longitude) using an EPSG OGC, this attribute would be urn:ogc:def:uom:EPSG::4236. A profile should limit this code to a set of allowed URNs agreed to by all sending and receiving parties.

(ABIE) DateAndOrTime

The Date and or the Time.

!# shows all attributes of DateAndOrTime.

Table !# - Attributes of CEEDS_ValidatedHistoricalData Document contextual model::DateAndOrTime

Attribute name	mult.	Attribute type	Description
dateTime (ABIE)	1..1	DateTime	Date and time as per ISO 8601 YYYY-MM-DDThh:mm:ss.sssZ.

(ABIE) EnvAsset

EnvAsset is an energy data space specific attribute referring to the Asset from which data is being requested

!# shows all attributes of EnvAsset.

Table !# - Attributes of CEEDS_ValidatedHistoricalData Document contextual model::EnvAsset

Attribute name	mult.	Attribute type	Description
type (ABIE)	0..1	String	type is an energy data space specific attribute that specifies type of asset refered to.
operatorId (ABIE)	0..1	String	operatorId is an energy data space specific attribute that specifies the id operator of the asset refered to.
meterId (ABIE)	0..1	String	meterId is an energy data space specific attribute that specifies the id of the meter that measures the asset.

(ABIE) Envelope

The envelope entity is a non-CIM object used in an energy data space environment to serve as a central top-level entity connecting the MarketDocumentHeader (energy data space specific context information block) with the regular CIM MarketDocument.

(ABIE) FlowDirection

The coded identification of the direction of energy flow.

!# shows all attributes of FlowDirection.

Table !# - Attributes of CEEDS_ValidatedHistoricalData Document contextual model::FlowDirection

Attribute name	mult.	Attribute type	Description
direction (ABIE)	1..1	DirectionKind_String	The coded identification of the direction of energy flow.

(ABIE) Fuel

A class indicating the origin of the fuel used at the related object.

!# shows all attributes of Fuel.

Table !# - Attributes of CEEDS_ValidatedHistoricalData Document contextual model::Fuel

Attribute name	mult.	Attribute type	Description
fuel (ABIE)	1..1	Fuel_String	An indication of the fuel used for the energy production, or part of the energy production, that is potentially fed into the grid at the related object.

(ABIE) Geometry

Geo-spatial representation of a physical location. A location may be represented as a Point, Line, Polygon, or Circle.

!# shows all attributes of Geometry.

Table !# - Attributes of CEEDS_ValidatedHistoricalData Document contextual model::Geometry

Attribute name	mult.	Attribute type	Description
identifier (ABIE)	0..1	UUID	A universally unique object identifier. Used to uniquely identify persistent objects between CIM messages.
mRID (ABIE)	0..1	String	Master resource identifier issued by a model authority. The mRID is unique within an exchange context. Global uniqueness is easily achieved by using a UUID, as specified in IETF RFC 4122, for the mRID. The use of UUID is strongly recommended. For CIMXML data files in RDF syntax conforming to IEC 61970-552, the mRID is mapped to rdf:ID or rdf:about attributes that identify CIM object elements.

!# shows all association ends of Geometry with other classes.

Table !# - Association ends of CEEDS_ValidatedHistoricalData Document contextual model::Geometry with other classes

mult.	mult.	Class type name	Description
PositionPoint (ABIE)	[0..1]	PositionPoint	Association Based On : ESMPClasses:: PositionPoint.PositionPoint[0..*] ----- CEEDS_ESMP_GridHostCap_Ext:: Geometry.[]
CoordinateSystem (ABIE)	[0..1]	CoordinateSystem	The coordinate system used for a given geometry. Association Based On : CEEDS_ESMP_GridHostCap_Ext:: Geometry.[] ----- ESMPClasses:: CoordinateSystem.CoordinateSystem[0..1]

(ABIE) Location

The place, scene, or point of something where someone or something has been, is, and/or will be at a given moment in time. It can be defined with one or more position points (coordinates) in a given coordinate system.

!# shows all attributes of Location.

Table !# - Attributes of CEEDS_ValidatedHistoricalData Document contextual model::Location

Attribute name	mult.	Attribute type	Description
mRID (ABIE)	0..1	String	The unique identification of a location. In the ESMP context, the "model authority" is defined as an authorized issuing office that provides an agreed identification coding scheme for market participant, domain, measurement point, resources (generator, lines, substations, etc.) identification, location, etc. Master resource identifier issued by a model authority. The mRID is globally unique within an exchange context. Global uniqueness is easily achieved by using a UUID for the mRID. It is strongly recommended to do this. For CIMXML data files in RDF syntax, the mRID is mapped to rdf:ID or rdf:about attributes that identify CIM object elements.
name (ABIE)	0..1	String	The name is any free human readable and possibly non unique text naming the object.

!# shows all association ends of Location with other classes.

Table !# - Association ends of CEEDS_ValidatedHistoricalData Document contextual model::Location with other classes

	mult.	mult.	Class type name	Description
Geometry (ABIE)	[0..1]		Geometry	The geometry for a given location. Association Based On : ESMPClasses:: Location.[] ----- CEEDS_ESMP_GridHostCap_Ext:: Geometry.Geometry[0..*]

(ABIE) MarketParticipant

The identification of the party participating in energy market business processes.

!# shows all attributes of MarketParticipant.

Table !# - Attributes of CEEDS_ValidatedHistoricalData Document contextual model::MarketParticipant

Attribute name	mult.	Attribute type	Description
mRID (ABIE)	1..1	PartyID_String	The identification of a party in the energy market. In the ESMP context, the "model authority" is defined as an authorized issuing office that provides an agreed identification coding scheme for market participant, domain, measurement point, resources (generator, lines, substations, etc.) identification. Master resource identifier issued by a model authority. The mRID is globally unique within an exchange context. Global uniqueness is easily achieved by using a UUID for the mRID. It is strongly recommended to do this. For CIMXML data files in RDF syntax, the mRID is mapped to rdf:ID or rdf:about attributes that identify CIM object elements.
name (ABIE)	0..1	String	The name is any free human readable and possibly non unique text naming the object.

!# shows all association ends of MarketParticipant with other classes.

Table !# - Association ends of CEEDS_ValidatedHistoricalData Document contextual model::MarketParticipant with other classes

	mult.	mult.	Class type name	Description
MarketRole (ABIE)	[1..1]		MarketRole	Association Based On : ESMPClasses:: MarketRole.MarketRole[1..1] ----- ESMPClasses:: MarketParticipant.[]

(ABIE) MarketRole

The identification of the intended behaviour of a market participant played within a given business process.

!# shows all attributes of MarketRole.

Table !# - Attributes of CEEDS_ValidatedHistoricalData Document contextual model::MarketRole

Attribute name	mult.	Attribute type	Description
type (ABIE)	1..1	MarketRoleKind_String	The identification of the role played by a market player.

(ABIE) Measure_Unit

The particular quantity, defined and adopted by convention, with which other quantities of the same kind are compared in order to express their magnitudes relative to that quantity.

!# shows all attributes of Measure_Unit.

Table !# - Attributes of CEEDS_ValidatedHistoricalData Document contextual model::Measure_Unit

Attribute name	mult.	Attribute type	Description
name (ABIE)	1..1	MeasurementUnitKind_String	The identification of the formal code for a measurement unit (UN/ECE Recommendation 20).

(ABIE) MessageDocumentHeader

MessageDocumentHeader is an energy data space specific entity with the creationDateTime of this document.

!# shows all attributes of MessageDocumentHeader.

Table !# - Attributes of CEEDS_ValidatedHistoricalData Document contextual model::MessageDocumentHeader

Attribute name	mult.	Attribute type	Description
creationDateTime (ABIE)	0..1	ESMP_DateTime	The date and time the MessageDocumenHeader was created.

!# shows all association ends of MessageDocumentHeader with other classes.

Table !# - Association ends of CEEDS_ValidatedHistoricalData Document contextual model::MessageDocumentHeader with other classes

mult.	mult.	Class type name	Description
MetaInformation (ABIE)	[0..1]	MetaInformation	Association Based On : CEEDS_ESMP_Envelope:: MessageDocumentHeader.[] ----- CEEDS_ESMP_Envelope:: MetaInformation.MetaInformation[0..1]

(ABIE) MetaInformation

MetaInformation is an energy data space specific entity referring to the information concerning the technical and functional context of the MarketDocument.

!# shows all attributes of MetaInformation.

Table !# - Attributes of CEEDS_ValidatedHistoricalData Document contextual model::MetaInformation

Attribute name	mult.	Attribute type	Description
connectionId (ABIE)	0..1	String	connectionId is an energy data space specific attribute referring to the established connection used for the communication of this document.
requestPermissionId (ABIE)	0..1	String	The requestPermissionId is energy data space specific identification attribute holding an unique identifier issued by the energy data space to allow for tracing of the permission through serveral message exchanges. Note: the requestPermissionId is NOT the identifier issued by the Identity Service Provider as specified in the EU Implementing Act on access to Metering and Consumption Data.
dataNeedId (ABIE)	0..1	String	dataNeedId is an energy data space specific attribute holding the type of data need requested, like â€˜last 24 hours at 1 hour intervalâ€ .
documentType (ABIE)	0..1	String	documentType is an energy data space specific attribute referring to the type of document, comparable to the MessageTypeList in the CIM part of the message. Note: This is applicable if you have the need to know the content without having to open the complete document. DocumentType might contain messages not (yet) part of the MessageTypeList.
finalCustomerId (ABIE)	0..1	String	The finalCustomerId is an energy data space specific attribute identifying the actor doing the request for data or data sharing.
dataSourceId (ABIE)	0..1	String	dataSourceId is an energy data space specific attribute referring to the datasource from which the requested data is being queried.
defaultValues (ABIE)	0..1	String	defaultValues is a energy data space specific attribute to hold relevant default values. The format to use can freely be chosen based on the implementation. Most commonly you would expect a semi-colon separated name value pair, like: "minPower=value; maxPower=value; avgPower=value".
regionConnector (ABIE)	0..1	String	regionConnector is an energy data space specific attribute that specifies the regional connector being used to retrieve the data requested.
regionCountry (ABIE)	0..1	String	regionCountry is an energy data space specific attribute that specifies the country where the regional connector is located.

!# shows all association ends of MetaInformation with other classes.

Table !# - Association ends of CEEDS_ValidatedHistoricalData Document contextual model::MetaInformation with other classes

mult.	mult.	Class type name	Description
EnvAsset (ABIE)	[0..1]	EnvAsset	Association Based On : CEEDS_ESMP_Envelope:: MetaInformation.[] ----- CEEDS_ESMP_Envelope:: EnvAsset.EnvAsset[0..1]

(ABIE) MeterReading

Set of values obtained from the meter.

!# shows all attributes of MeterReading.

Table !# - Attributes of CEEDS_ValidatedHistoricalData Document contextual model::MeterReading

Attribute	mult.	Attribute type	Description

name

			The unique identification of the meter reading. In the ESMP context, the "model authority" is defined as a party (originator of the exchange) that provides an identification in the context of a business exchange.
mRID (ABIE)	0..1	ResourceID_String	Master resource identifier issued by a model authority. The mRID is unique within an exchange context. Global uniqueness is easily achieved by using a UUID, as specified in RFC 4122, for the mRID. The use of UUID is strongly recommended. For CIMXML data files in RDF syntax conforming to IEC 61970-552 Edition 1, the mRID is mapped to rdf:ID or rdf:about attributes that identify CIM object elements.
name (ABIE)	0..1	String	The name is any free human readable and possibly non unique text naming the object.

!# shows all association ends of MeterReading with other classes.

Table !# - Association ends of CEEDS_ValidatedHistoricalData Document contextual model::MeterReading with other classes

mult.	mult.	Class type name	Description
Readings (ABIE)	[0..1]	Reading	All reading values contained within this meter reading. Association Based On : ESMPClasses:: Reading.Readings[0..*] ----- ESMPClasses:: MeterReading.[]

(ABIE) MktPSRType

The type of a power system resource.

!# shows all attributes of MktPSRType.

Table !# - Attributes of CEEDS_ValidatedHistoricalData Document contextual model::MktPSRType

Attribute name	mult.	Attribute type	Description
psrType (ABIE)	1..1	PsrType_String	The coded type of a power system resource.

(ABIE) Point

The identification of the values being addressed within a specific interval of time.

!# shows all attributes of Point.

Table !# - Attributes of CEEDS_ValidatedHistoricalData Document contextual model::Point

Attribute name	mult.	Attribute type	Description
position (ABIE)	1..1	Position_Integer	A sequential value representing the relative position within a given time interval.

!# shows all association ends of Point with other classes.

Table !# - Association ends of CEEDS_ValidatedHistoricalData Document contextual model::Point with other classes

mult.	mult.	Class type name	Description
EnergyQuality_Quantity (ABIE)	[0..1]	Quantity	The Quantity information associated with a given Point. Association Based On : ESMPClasses:: Quantity.Quantity[0..*] ----- ESMPClasses:: Point.[]
Energy_Quantity (ABIE)	[0..1]	Quantity	The Quantity information associated with a given Point. Association Based On : ESMPClasses:: Quantity.Quantity[0..*] ----- ESMPClasses:: Point.[]

(ABIE) PositionPoint

Set of spatial coordinates that determine a point, defined in the coordinate system specified in 'Location.CoordinateSystem'. Use a single position point instance to describe a point-oriented location. Use a sequence of position points to describe a line-oriented object (physical location of non-point oriented objects like cables or lines), or area of an object (like a substation or a geographical zone - in this case, have first and last position point with the same values).

!# shows all attributes of PositionPoint.

Table !# - Attributes of CEEDS_ValidatedHistoricalData Document contextual model::PositionPoint

Attribute name	mult.	Attribute type	Description
xPosition (ABIE)	0..1	String	X axis position.
yPosition (ABIE)	0..1	String	Y axis position.
zPosition (ABIE)	0..1	String	(if applicable) Z axis position.

(ABIE) Process

The formal identification of the business process in which a flow of information is exchanged.

!# shows all attributes of Process.

Table !# - Attributes of CEEDS_ValidatedHistoricalData Document contextual model::Process

Attribute name	mult.	Attribute type	Description
processType (ABIE)	1..1	ProcessKind_String	The identification of the nature of process that the document addresses.

(ABIE) Quantity

Description of quantities needed in the data exchange.
The type of the quantity is described either by the role of the association or the quantityType attribute.
The quality attribute provides the information about the quality of the quantity (measured, estimated, etc.).

!# shows all attributes of Quantity.

Table !# - Attributes of CEEDS_ValidatedHistoricalData Document contextual model::Quantity

Attribute name	mult.	Attribute type	Description
quantity (ABIE)	1..1	Decimal	The quantity value. The association role provides the information about what is expressed.
quality (ABIE)	0..1	Quality_String	The description of the quality of the quantity.
type (ABIE)	0..1	String	The description of the type of the quantity.

!# shows all association ends of Quantity with other classes.

Table !# - Association ends of CEEDS_ValidatedHistoricalData Document contextual model::Quantity with other classes

mult.	mult.	Class type name	Description
QuantityDefinition (ABIE)	[1..1]	QuantityDefinition	Association Based On : CEEDS_ESMP_Activation:: QuantityDefinition.QuantityDefinition[1..1] ----- ESMPClasses:: Quantity.[]

(ABIE) QuantityDefinition

Defines the characteristics of the quantities included in TimeSeries. QuantityDefinitions may be defined and referenced by the Quantity instances to further describe the Quantity values.

!# shows all attributes of QuantityDefinition.

Table !# - Attributes of CEEDS_ValidatedHistoricalData Document contextual model::QuantityDefinition

Attribute name	mult.	Attribute type	Description
mRID (ABIE)	0..1	String	Master resource identifier issued by a model authority. The mRID is unique within an exchange context. Global uniqueness is easily achieved by using a UUID, as specified in IETF RFC 4122, for the mRID. The use of UUID is strongly recommended. For CIMXML data files in RDF syntax conforming to IEC 61970-552, the mRID is mapped to rdf:ID or rdf:about attributes that identify CIM object elements.
quantityType (ABIE)	0..1	String	Describes an observable property that can be measured and quantified numerically. Electricity related examples are ActivePower, ReactivePower, ActiveEnergy, ReactiveEnergy.
commodityFlowOccurence (ABIE)	0..1	CommodityDirectionKind	The direction of flow, such as that of water or electricity, for the quantity. Quantities can be for measurements, or expected future values, e.g. forecast, schedule, instruction.
quantityQualifier (ABIE)	0..1	String	Role or characteristic of the quantity type. For example: average measured value, minimum measured value, maximum measured value, forecast expected value, forecast lower bound, forecast upper bound, lower operational limit, upper operational limit.

!# shows all association ends of QuantityDefinition with other classes.

Table !# - Association ends of CEEDS_ValidatedHistoricalData Document contextual model::QuantityDefinition with other classes

	mult.	mult.	Class type name	Description
Measurement_Unit (ABIE)	[1..1]		Measure_Unit	Association Based On : CEEDS_ESMP_Activation:: QuantityDefinition.[] ----- ESMPClasses:: Measure_Unit.Unit[1..1]

(ABIE) Reading

Specific value measured by a meter or other asset, or calculated by a system. Each Reading is associated with a specific ReadingType.

!# shows all attributes of Reading.

Table !# - Attributes of CEEDS_ValidatedHistoricalData Document contextual model::Reading

Attribute name	mult.	Attribute type	Description
mRID (ABIE)	0..1	ResourceID_String	The identification of the reading. Master resource identifier issued by a model authority. The mRID is unique within an exchange context. Global uniqueness is easily achieved by using a UUID, as specified in RFC 4122, for the mRID. The use of UUID is strongly recommended. For CIMXML data files in RDF syntax conforming to IEC 61970-552 Edition 1, the mRID is mapped to rdf:ID or rdf:about attributes that identify CIM object elements.

!# shows all association ends of Reading with other classes.

Table !# - Association ends of CEEDS_ValidatedHistoricalData Document contextual model::Reading with other classes

	mult.	mult.	Class type name	Description
ReadingType (ABIE)	[0..1]		ReadingType	Type information for this reading value. Association Based On : ESMPClasses:: Reading.[] ----- CEEDS_ESMP_ReadingType:: ReadingType.ReadingType[1]

(ABIE) ReadingType

Detailed description for a type of a reading value. Values in attributes allow for the creation of recommended codes to be used for identifying reading value types as follows: .....

!# shows all attributes of ReadingType.

Table !# - Attributes of CEEDS_ValidatedHistoricalData Document contextual model::ReadingType

Attribute name	mult.	Attribute type	Description
aggregate (ABIE)	0..1	AggregateKind	Salient attribute of the reading data aggregated from individual endpoints. This is mainly used to define a mathematical operation carried out over 'macroPeriod', but may also be used to describe an attribute of the data when the 'macroPeriod' is not defined.
accumulation (ABIE)	0..1	AccumulationKind	Accumulation behaviour of a reading over time, usually 'measuringPeriod', to be used with individual endpoints (as opposed to 'macroPeriod' and 'aggregate' that are used to describe aggregations of data from individual endpoints).
commodity (ABIE)	0..1	CommodityKind	Commodity being measured.

(ABIE) Reason

The motivation of an act.

!# shows all attributes of Reason.

Table !# - Attributes of CEEDS_ValidatedHistoricalData Document contextual model::Reason

Attribute name	mult.	Attribute type	Description
code (ABIE)	1..1	ReasonCode_String	The motivation of an act in coded form.
text (ABIE)	0..1	ReasonText_String	The textual explanation corresponding to the reason code.

(ABIE) RegisteredResource

A resource that is registered through the market participant registration system. Examples include generating unit, load, and non-physical

generator or load.

!# shows all attributes of RegisteredResource.

Table !# - Attributes of CEEDS_ValidatedHistoricalData Document contextual model::RegisteredResource

Attribute name	mult.	Attribute type	Description
mRID (ABIE)	1..1	ResourceID_String	The unique identification of a resource. In the ESMP context, the "model authority" is defined as an authorized issuing office that provides an agreed identification coding scheme for market participant, domain, measurement point, resources (generator, lines, substations, etc.) identification. Master resource identifier issued by a model authority. The mRID is globally unique within an exchange context. Global uniqueness is easily achieved by using a UUID for the mRID. It is strongly recommended to do this. For CIMXML data files in RDF syntax, the mRID is mapped to rdf:ID or rdf:about attributes that identify CIM object elements.
name (ABIE)	0..1	String	The name is any free human readable and possibly non unique text naming the object.
description (ABIE)	0..1	String	The description is a free human readable text describing or naming the object. It may be non unique and may not correlate to a naming hierarchy.

!# shows all association ends of RegisteredResource with other classes.

Table !# - Association ends of CEEDS_ValidatedHistoricalData Document contextual model::RegisteredResource with other classes

mult.	mult.	Class type name	Description
PSRType (ABIE)	[0..1]	MktPSRType	The identification of the type of resource associated with this RegisteredResource. Association Based On : ESMPClasses:: MktPSRType.PSRType[0..1] ----- ESMPClasses:: RegisteredResource.[]
Location (ABIE)	[0..1]	Location	Location of this power system resource. Association Based On : ESMPClasses:: Location.Location[0..1] ----- ESMPClasses:: RegisteredResource.[]
Fuel (ABIE)	[0..1]	Fuel	Association Based On : ESMPClasses:: Fuel.Fuel[0..*] ----- ESMPClasses:: RegisteredResource.[]

(ABIE) Series_Period

The identification of the period of time corresponding to a given time interval and resolution.

!# shows all attributes of Series_Period.

Table !# - Attributes of CEEDS_ValidatedHistoricalData Document contextual model::Series_Period

Attribute name	mult.	Attribute type	Description
resolution (ABIE)	1..1	Duration	The definition of the number of units of time that compose an individual step within a period.
timeInterval (ABIE)	1..1	ESMP_DateTimeInterval	The start and end time of the period.

!# shows all association ends of Series_Period with other classes.

Table !# - Association ends of CEEDS_ValidatedHistoricalData Document contextual model::Series_Period with other classes

mult.	mult.	Class type name	Description
Reason (ABIE)	[0..1]	Reason	The reason information associated with a Series_Period providing motivation information. Association Based On : ESMPClasses:: Reason.Reason[0..*] ----- ESMPClasses:: Series_Period.[]

(ABIE) TimeSeries

A set of time-ordered quantities being exchanged in relation to a product.
In the ESMP profile, the TimeSeries provides not only time-ordered quantities but also time-ordered information.

!# shows all attributes of TimeSeries.

Table !# - Attributes of CEEDS_ValidatedHistoricalData Document contextual model::TimeSeries

Attribute name	mult.	Attribute type	Description
mRID (ABIE)	0..1	ID_String	A unique identification of the time series. In the ESMP context, the "model authority" is defined as a party (originator of the exchange) that provides a unique identification in the context of a business exchange such as time series identification, bid identification, ... Master resource identifier issued by a model authority. The mRID is globally unique within an exchange context. Global uniqueness is easily achieved by using a UUID for the mRID. It is strongly recommended to do this. For CIMXML data files in RDF syntax, the mRID is mapped to rdf:ID or rdf:about attributes that identify CIM object elements.
version (ABIE)	1..1	ESMPVersion_String	The identification of the version of the time series.
businessType (ABIE)	0..1	BusinessKind_String	The identification of the nature of the time series.
product (ABIE)	0..1	EnergyProductKind_String	The identification of the nature of an energy product such as power, energy, reactive power, etc.

!# shows all association ends of TimeSeries with other classes.

Table !# - Association ends of CEEDS_ValidatedHistoricalData Document contextual model::TimeSeries with other classes

mult.	mult.	Class type name	Description
RegisteredResource (ABIE)	[0..1]	RegisteredResource	The identification of a resource associated with a TimeSeries. Association Based On : ESMPClasses:: RegisteredResource.RegisteredResource[0..*] ----- ESMPClasses:: TimeSeries.[]
Reason (ABIE)	[0..1]	Reason	The reason information associated with a TimeSeries providing motivation information. Association Based On : ESMPClasses:: TimeSeries.[] ----- ESMPClasses:: Reason.Reason[0..*]
Energy_Measurement_Unit (ABIE)	[0..1]	Measure_Unit	The unit of measure associated with the quantities in a TimeSeries. Association Based On : ESMPClasses:: TimeSeries.[] ----- ESMPClasses:: Measure_Unit.Measurement_Unit[0..*]
EnergyQuality_Measurement_Unit (ABIE)	[0..1]	Measure_Unit	The unit of measure associated with the quantities in a TimeSeries. Association Based On : ESMPClasses:: TimeSeries.[] ----- ESMPClasses:: Measure_Unit.Measurement_Unit[0..*]
FlowDirection (ABIE)	[1..1]	FlowDirection	The flow direction associated with a TimeSeries. Association Based On : ESMPClasses:: TimeSeries.[] ----- ESMPClasses:: FlowDirection.FlowDirection[0..1]
DateAndOrTime (ABIE)	[1..1]	DateAndOrTime	A date and/or time associated with a TimeSeries. Association Based On : ESMPClasses:: TimeSeries.[] ----- ESMPClasses:: DateAndOrTime.DateAndOrTime[0..*]
AccountingPoint (ABIE)	[0..1]	AccountingPoint	The identification of a measurement point associated with a TimeSeries. Association Based On : ESMPClasses:: TimeSeries.[] -----

ESMPClasses:: MarketEvaluationPoint.MarketEvaluationPoint[0..*]

(ACC) UsagePointLocation

Location of an individual usage point.

!# shows all attributes of UsagePointLocation.

Table !# - Attributes of CEEDS_ValidatedHistoricalData Document contextual model::UsagePointLocation

Attribute name	mult.	Attribute type	Description
geoInfoReference (ACC)	1..1	String	(if applicable) Reference to geographical information source, often external to the utility.

(ABIE) VHD_MarketDocument

An electronic document containing the information necessary to satisfy the requirements of a given business process.

!# shows all attributes of VHD_MarketDocument.

Table !# - Attributes of CEEDS_ValidatedHistoricalData Document contextual model::VHD_MarketDocument

Attribute name	mult.	Attribute type	Description
mRID (ABIE)	1..1	ID_String	The unique identification of the document being exchanged within a business process flow. In the ESMP context, the "model authority" is defined as a party (originator of the exchange) that provides an identification in the context of a business exchange such as document identification, ... Master resource identifier issued by a model authority. The mRID is globally unique within an exchange context. Global uniqueness is easily achieved by using a UUID for the mRID. It is strongly recommended to do this. For CIMXML data files in RDF syntax, the mRID is mapped to rdf:ID or rdf:about attributes that identify CIM object elements.
createdDateTime (ABIE)	1..1	ESMP_DateTime	The date and time of the creation of the document.
description (ABIE)	0..1	String	The description is a free human readable text describing or naming the object. It may be non unique and may not correlate to a naming hierarchy.
revisionNumber (ABIE)	0..1	ESMPVersion_String	The identification of the version that distinguishes one evolution of a document from another.
type (ABIE)	0..1	MessageKind_String	The coded type of a document. The document type describes the principal characteristic of the document.

!# shows all association ends of VHD_MarketDocument with other classes.

Table !# - Association ends of CEEDS_ValidatedHistoricalData Document contextual model::VHD_MarketDocument with other classes

mult.	mult.	Class type name	Description
Process (ABIE)	[1..1]	Process	The Process associated with an electronic document header that is valid for the whole document. Association Based On : ESMPClasses:: MarketDocument.[] ----- ESMPClasses:: Process.Process[0..*]
Receiver_MarketParticipant (ABIE)	[1..1]	MarketParticipant	Association Based On : ESMPClasses:: MarketParticipant.MarketParticipant[0..1] ----- ESMPClasses:: MarketDocument.[]
Sender_MarketParticipant (ABIE)	[1..1]	MarketParticipant	Association Based On : ESMPClasses:: MarketParticipant.MarketParticipant[0..1] ----- ESMPClasses:: MarketDocument.[]

CEEDS_ValidatedHistoricalData Document assembly model

General

!# shows the diagram.

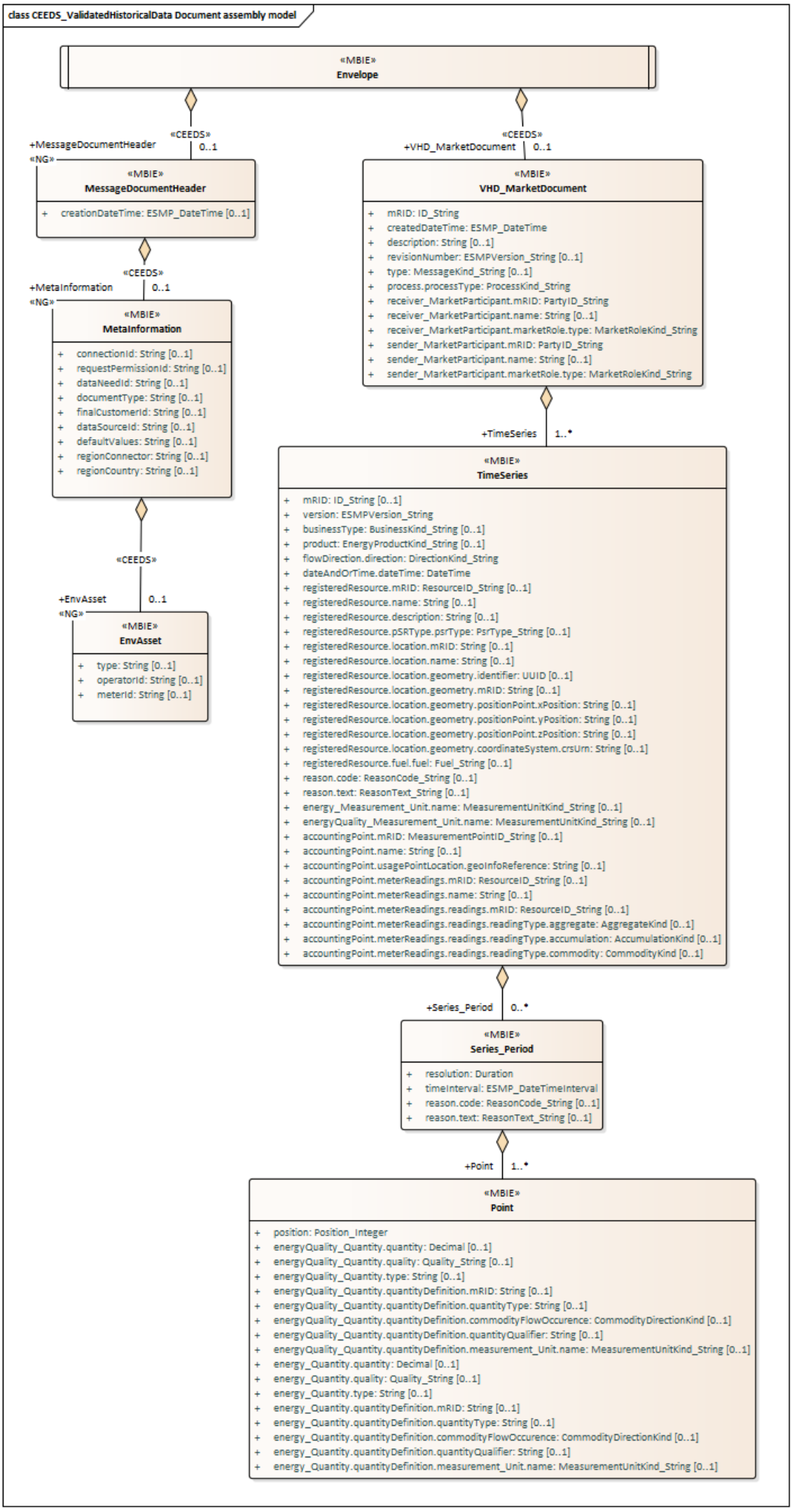

class CEEDS_ValidatedHistoricalData Document assembly model
«MBIE»
Envelope
«CEEDS»
+MessageDocumentHeader
0..1
«NG»
«MBIE»
MessageDocumentHeader
+ creationDateTime: ESMP_DateTime [0..1]
«CEEDS»
+MetaInformation
0..1
«NG»
«MBIE»
MetaInformation
+ connectionId: String [0..1]
+ requestPermissionId: String [0..1]
+ dataNeedId: String [0..1]
+ documentType: String [0..1]
+ finalCustomerId: String [0..1]
+ dataSourceId: String [0..1]
+ defaultValues: String [0..1]
+ regionConnector: String [0..1]
+ regionCountry: String [0..1]
«CEEDS»
+EnvAsset
0..1
«NG»
«MBIE»
EnvAsset
+ type: String [0..1]
+ operatorId: String [0..1]
+ meterId: String [0..1]
«CEEDS»
+VHD_MarketDocument
0..1
«MBIE»
VHD_MarketDocument
+ mRID: ID_String
+ createdDateTime: ESMP_DateTime
+ description: String [0..1]
+ revisionNumber: ESMPVersion_String [0..1]
+ type: MessageKind_String [0..1]
+ process.processType: ProcessKind_String
+ receiver_MarketParticipant.mRID: PartyID_String
+ receiver_MarketParticipant.name: String [0..1]
+ receiver_MarketParticipant.marketRole.type: MarketRoleKind_String
+ sender_MarketParticipant.mRID: PartyID_String
+ sender_MarketParticipant.name: String [0..1]
+ sender_MarketParticipant.marketRole.type: MarketRoleKind_String
+TimeSeries
1..*
«MBIE»
TimeSeries
+ mRID: ID_String [0..1]
+ version: ESMPVersion_String
+ businessType: BusinessKind_String [0..1]
+ product: EnergyProductKind_String [0..1]
+ flowDirection.direction: DirectionKind_String
+ dateAndOrTime.dateTime: DateTime
+ registeredResource.mRID: ResourceID_String [0..1]
+ registeredResource.name: String [0..1]
+ registeredResource.description: String [0..1]
+ registeredResource.pSRType.psrType: PsrType_String [0..1]
+ registeredResource.location.mRID: String [0..1]
+ registeredResource.location.name: String [0..1]
+ registeredResource.location.geometry.identifier: UUID [0..1]
+ registeredResource.location.geometry.mRID: String [0..1]
+ registeredResource.location.geometry.positionPoint.xPosition: String [0..1]
+ registeredResource.location.geometry.positionPoint.yPosition: String [0..1]
+ registeredResource.location.geometry.positionPoint.zPosition: String [0..1]
+ registeredResource.location.geometry.coordinateSystem.crsUrn: String [0..1]
+ registeredResource.fuel.fuel: Fuel_String [0..1]
+ reason.code: ReasonCode_String [0..1]
+ reason.text: ReasonText_String [0..1]
+ energy_Measurement_Unit.name: MeasurementUnitKind_String [0..1]
+ energyQuality_Measurement_Unit.name: MeasurementUnitKind_String [0..1]
+ accountingPoint.mRID: MeasurementPointID_String [0..1]
+ accountingPoint.name: String [0..1]
+ accountingPoint.usagePointLocation.geoInfoReference: String [0..1]
+ accountingPoint.meterReadings.mRID: ResourceID_String [0..1]
+ accountingPoint.meterReadings.name: String [0..1]
+ accountingPoint.meterReadings.readings.mRID: ResourceID_String [0..1]
+ accountingPoint.meterReadings.readings.readingType.aggregate: AggregateKind [0..1]
+ accountingPoint.meterReadings.readings.readingType.accumulation: AccumulationKind [0..1]
+ accountingPoint.meterReadings.readings.readingType.commodity: CommodityKind [0..1]
+Series_Period
0..*
«MBIE»
Series_Period
+ resolution: Duration
+ timeInterval: ESMP_DateTimeInterval
+ reason.code: ReasonCode_String [0..1]
+ reason.text: ReasonText_String [0..1]
+Point
1..*
«MBIE»
Point
+ position: Position_Integer
+ energyQuality_Quantity.quantity: Decimal [0..1]
+ energyQuality_Quantity.quality: Quality_String [0..1]
+ energyQuality_Quantity.type: String [0..1]
+ energyQuality_Quantity.quantityDefinition.mRID: String [0..1]
+ energyQuality_Quantity.quantityDefinition.quantityType: String [0..1]
+ energyQuality_Quantity.quantityDefinition.commodityFlowOccurence: CommodityDirectionKind [0..1]
+ energyQuality_Quantity.quantityDefinition.quantityQualifier: String [0..1]
+ energyQuality_Quantity.quantityDefinition.measurement_Unit.name: MeasurementUnitKind_String [0..1]
+ energy_Quantity.quantity: Decimal [0..1]
+ energy_Quantity.quality: Quality_String [0..1]
+ energy_Quantity.type: String [0..1]
+ energy_Quantity.quantityDefinition.mRID: String [0..1]
+ energy_Quantity.quantityDefinition.quantityType: String [0..1]
+ energy_Quantity.quantityDefinition.commodityFlowOccurence: CommodityDirectionKind [0..1]
+ energy_Quantity.quantityDefinition.quantityQualifier: String [0..1]
+ energy_Quantity.quantityDefinition.measurement_Unit.name: MeasurementUnitKind_String [0..1]

Figure !# - diagram CEEDS_ValidatedHistoricalData Document assembly model

(MBIE) EnvAsset

EnvAsset is an energy data space specific attribute referring to the Asset from which data is being requested

!# shows all attributes of EnvAsset.

Table !# - Attributes of CEEDS_ValidatedHistoricalData Document assembly model::EnvAsset

Attribute name	mult.	Attribute type	Description
type (MBIE)	0..1	String	type is an energy data space specific attribute that specifies type of asset refered to.
operatorId (MBIE)	0..1	String	operatorId is an energy data space specific attribute that specifies the id operator of the asset refered to.
meterId (MBIE)	0..1	String	meterId is an energy data space specific attribute that specifies the id of the meter that measures the asset.

(MBIE) Envelope

The envelope entity is a non-CIM object used in an energy data space environment to serve as a central top-level entity connecting the MarketDocumentHeader (energy data space specific context information block) with the regular CIM MarketDocument.

(MBIE) MessageDocumentHeader

MessageDocumentHeader is an energy data space specific entity with the creationDateTime of this document.

!# shows all attributes of MessageDocumentHeader.

Table !# - Attributes of CEEDS_ValidatedHistoricalData Document assembly model::MessageDocumentHeader

Attribute name	mult.	Attribute type	Description
creationDateTime (MBIE)	0..1	ESMP_DateTime	The date and time the MessageDocumenHeader was created.

!# shows all association ends of MessageDocumentHeader with other classes.

Table !# - Association ends of CEEDS_ValidatedHistoricalData Document assembly model::MessageDocumentHeader with other classes

mult.	mult.	Class type name	Description
MetaInformation (MBIE)	[0..1]	MetaInformation	Association Based On : CEEDS_ValidatedHistoricalData Document contextual model:: MessageDocumentHeader.[] ----- CEEDS_ValidatedHistoricalData Document contextual model:: MetaInformation.MetaInformation[0..1]

(MBIE) MetaInformation

MetaInformation is an energy data space specific entity referring to the information concerning the technical and functional context of the MarketDocument.

!# shows all attributes of MetaInformation.

Table !# - Attributes of CEEDS_ValidatedHistoricalData Document assembly model::MetaInformation

Attribute name	mult.	Attribute type	Description
connectionId (MBIE)	0..1	String	connectionId is an energy data space specific attribute referring to the established connection used for the communication of this document.
requestPermissionId (MBIE)	0..1	String	The requestPermissionId is energy data space specific identification attribute holding an unique identifier issued by the energy data space to allow for tracing of the permission through serveral message exchanges. Note: the requestPermissionId is NOT the identifier issued by the Identity Service Provider as specified in the EU Implementing Act on access to Metering and Consumption Data.
dataNeedId (MBIE)	0..1	String	dataNeedId is an energy data space specific attribute holding the type of data need requested, like â€˜last 24 hours at 1 hour intervalâ€ .
documentType (MBIE)	0..1	String	documentType is an energy data space specific attribute referring to the type of document, comparable to the MessageTypeList in the CIM part of the message. Note: This is applicable if you have the need to know the content without having to open the complete document. DocumentType might contain messages not (yet) part of the MessageTypeList.
finalCustomerId (MBIE)	0..1	String	The finalCustomerId is an energy data space specific attribute identifying the actor doing the request for data or data sharing.
dataSourceId (MBIE)	0..1	String	dataSourceId is an energy data space specific attribute referring to the datasource from which the requested data is being queried.
defaultValues (MBIE)	0..1	String	defaultValues is a energy data space specific attribute to hold relevant default values. The format to use can freely be chosen based on the implementation. Most commonly you would expect a semi-colon separated name value pair, like: "minPower=value; maxPower=value; avgPower=value".
regionConnector (MBIE)	0..1	String	regionConnector is an energy data space specific attribute that specifies the regional connector being used to retrieve the data requested.
regionCountry (MBIE)	0..1	String	regionCountry is an energy data space specific attribute that specifies the country where the regional connector is located.

!# shows all association ends of MetaInformation with other classes.

Table !# - Association ends of CEEDS_ValidatedHistoricalData Document assembly model::MetaInformation with other classes

mult.	mult.	Class type name	Description
EnvAsset (MBIE)	[0..1]	EnvAsset	Association Based On : CEEDS_ValidatedHistoricalData Document contextual model:: MetaInformation.[] ----- CEEDS_ValidatedHistoricalData Document contextual model:: EnvAsset.EnvAsset[0..1]

(MBIE) Point

The identification of the values being addressed within a specific interval of time.

!# shows all attributes of Point.

Table !# - Attributes of CEEDS_ValidatedHistoricalData Document assembly model::Point

Attribute name	mult.	Attribute type	Description

position (MBIE)	1..1 Position_Integer	A sequential value representing the relative position within a given time interval.
energyQuality_Quantity.quantity (MBIE)	0..1 Decimal	The quantity value. The association role provides the information about what is expressed. --- The Quantity information associated with a given Point.
energyQuality_Quantity.quality (MBIE)	0..1 Quality_String	The description of the quality of the quantity. --- The Quantity information associated with a given Point.
energyQuality_Quantity.type (MBIE)	0..1 String	The description of the type of the quantity. --- The Quantity information associated with a given Point.
energyQuality_Quantity.quantityDefinition.mRID (MBIE)	0..1 String	Master resource identifier issued by a model authority. The mRID is unique within an exchange context. Global uniqueness is easily achieved by using a UUID, as specified in IETF RFC 4122, for the mRID. The use of UUID is strongly recommended. For CIMXML data files in RDF syntax conforming to IEC 61970-552, the mRID is mapped to rdf:ID or rdf:about attributes that identify CIM object elements. --- The Quantity information associated with a given Point.
energyQuality_Quantity.quantityDefinition.quantityType (MBIE)	0..1 String	Describes an observable property that can be measured and quantified numerically. Electricity related examples are ActivePower, ReactivePower, ActiveEnergy, ReactiveEnergy. --- The Quantity information associated with a given Point.
energyQuality_Quantity.quantityDefinition.commodityFlowOccurence (MBIE)	0..1 CommodityDirectionKind	The direction of flow, such as that of water or electricity, for the quantity. Quantities can be for measurements, or expected future values, e.g. forecast, schedule, instruction. --- The Quantity information associated with a given Point.
energyQuality_Quantity.quantityDefinition.quantityQualifier (MBIE)	0..1 String	Role or characteristic of the quantity type. For example: average measured value, minimum measured value, maximum measured value, forecast expected value, forecast lower bound, forecast upper bound, lower operational limit, upper operational limit. --- The Quantity information associated with a given

			Point.
energyQuality_Quantity.quantityDefinition.measurement_Unit.name (MBIE)	0..1	MeasurementUnitKind_String	The identification of the formal code for a measurement unit (UN/ECE Recommendation 20). --- The Quantity information associated with a given Point.
energy_Quantity.quantity (MBIE)	0..1	Decimal	The quantity value. The association role provides the information about what is expressed. --- The Quantity information associated with a given Point.
energy_Quantity.quality (MBIE)	0..1	Quality_String	The description of the quality of the quantity. --- The Quantity information associated with a given Point.
energy_Quantity.type (MBIE)	0..1	String	The description of the type of the quantity. --- The Quantity information associated with a given Point.
energy_Quantity.quantityDefinition.mRID (MBIE)	0..1	String	Master resource identifier issued by a model authority. The mRID is unique within an exchange context. Global uniqueness is easily achieved by using a UUID, as specified in IETF RFC 4122, for the mRID. The use of UUID is strongly recommended. For CIMXML data files in RDF syntax conforming to IEC 61970-552, the mRID is mapped to rdf:ID or rdf:about attributes that identify CIM object elements. --- The Quantity information associated with a given Point.
energy_Quantity.quantityDefinition.quantityType (MBIE)	0..1	String	Describes an observable property that can be measured and quantified numerically. Electricity related examples are ActivePower, ReactivePower, ActiveEnergy, ReactiveEnergy. --- The Quantity information associated with a given Point.
energy_Quantity.quantityDefinition.commodityFlowOccurence (MBIE)	0..1	CommodityDirectionKind	The direction of flow, such as that of water or electricity, for the quantity. Quantities can be for measurements,or expected future values, e.g. forecast, schedule, instruction. --- The Quantity information associated with a given Point.
energy_Quantity.quantityDefinition.quantityQualifier (MBIE)	0..1	String	Role or characteristic of the quantity type. For example: average measured value, minimum measured value, maximum measured value, forecast expected value, forecast lower bound,

			forecast upper bound, lower operational limit, upper operational limit. --- The Quantity information associated with a given Point.
energy_Quantity.quantityDefinition.measurement_Unit.name (MBIE)	0..1	MeasurementUnitKind_String	The identification of the formal code for a measurement unit (UN/ECE Recommendation 20). --- The Quantity information associated with a given Point.

(MBIE) Series_Period

The identification of the period of time corresponding to a given time interval and resolution.

!# shows all attributes of Series_Period.

Table !# - Attributes of CEEDS_ValidatedHistoricalData Document assembly model::Series_Period

Attribute name	mult.	Attribute type	Description
resolution (MBIE)	1..1	Duration	The definition of the number of units of time that compose an individual step within a period.
timeInterval (MBIE)	1..1	ESMP_DateTimeInterval	The start and end time of the period.
reason.code (MBIE)	0..1	ReasonCode_String	The motivation of an act in coded form. --- The reason information associated with a Series_Period providing motivation information.
reason.text (MBIE)	0..1	ReasonText_String	The textual explanation corresponding to the reason code. --- The reason information associated with a Series_Period providing motivation information.

(MBIE) TimeSeries

A set of time-ordered quantities being exchanged in relation to a product.
In the ESMP profile, the TimeSeries provides not only time-ordered quantities but also time-ordered information.

!# shows all attributes of TimeSeries.

Table !# - Attributes of CEEDS_ValidatedHistoricalData Document assembly model::TimeSeries

Attribute name	mult.	Attribute type	Description
mRID (MBIE)	0..1	ID_String	A unique identification of the time series. In the ESMP context, the "model authority" is defined as a party (originator of the exchange) that provides a unique identification in the context of a business exchange such as time series identification, bid identification, ... Master resource identifier issued by a model authority. The mRID is globally unique within an exchange context. Global uniqueness is easily achieved by using a UUID for the mRID. It is strongly recommended to do this. For CIMXML data files in RDF syntax, the mRID is mapped to rdf:ID or rdf:about attributes that identify CIM object elements.
version (MBIE)	1..1	ESMPVersion_String	The identification of the version of the time series.
			The identification of the nature

businessType (MBIE)	0..1 BusinessKind_String	of the time series.
product (MBIE)	0..1 EnergyProductKind_String	The identification of the nature of an energy product such as power, energy, reactive power, etc.
flowDirection.direction (MBIE)	1..1 DirectionKind_String	The coded identification of the direction of energy flow. --- The flow direction associated with a TimeSeries.
dateAndOrTime.dateTime (MBIE)	1..1 DateTime	Date and time as per ISO 8601 YYYY-MM-DDThh:mm:ss.sssZ. --- A date and/or time associated with a TimeSeries.
registeredResource.mRID (MBIE)	0..1 ResourceID_String	The unique identification of a resource. In the ESMP context, the "model authority" is defined as an authorized issuing office that provides an agreed identification coding scheme for market participant, domain, measurement point, resources (generator, lines, substations, etc.) identification. Master resource identifier issued by a model authority. The mRID is globally unique within an exchange context. Global uniqueness is easily achieved by using a UUID for the mRID. It is strongly recommended to do this. For CIMXML data files in RDF syntax, the mRID is mapped to rdf:ID or rdf:about attributes that identify CIM object elements. --- The identification of a resource associated with a TimeSeries.
registeredResource.name (MBIE)	0..1 String	The name is any free human readable and possibly non unique text naming the object. --- The identification of a resource associated with a TimeSeries.
registeredResource.description (MBIE)	0..1 String	The description is a free human readable text describing or naming the object. It may be non unique and may not correlate to a naming hierarchy. --- The identification of a resource associated with a TimeSeries.
registeredResource.pSRType.psrType (MBIE)	0..1 PsrType_String	The coded type of a power system resource. --- The identification of a resource associated with a TimeSeries. --- The identification of the type of resource associated with this RegisteredResource.
		The unique identification of a location. In the ESMP context, the

		"model authority" is defined as an authorized issuing office that provides an agreed identification coding scheme for market participant, domain, measurement point, resources (generator, lines, substations, etc.) identification, location, etc.
registeredResource.location.mRID (MBIE)	0..1 String	Master resource identifier issued by a model authority. The mRID is globally unique within an exchange context. Global uniqueness is easily achieved by using a UUID for the mRID. It is strongly recommended to do this. For CIMXML data files in RDF syntax, the mRID is mapped to rdf:ID or rdf:about attributes that identify CIM object elements.
registeredResource.location.name (MBIE)	0..1 String	--- The identification of a resource associated with a TimeSeries. --- Location of this power system resource. The name is any free human readable and possibly non unique text naming the object.
registeredResource.location.geometry.identifier (MBIE)	0..1 UUID	--- The identification of a resource associated with a TimeSeries. --- Location of this power system resource. --- The geometry for a given location. A universally unique object identifier. Used to uniquely identify persistent objects between CIM messages.
registeredResource.location.geometry.mRID (MBIE)	0..1 String	--- The identification of a resource associated with a TimeSeries. --- Location of this power system resource. --- The geometry for a given location. Master resource identifier issued by a model authority. The mRID is unique within an exchange context. Global uniqueness is easily achieved by using a UUID, as specified in IETF RFC 4122, for the mRID. The use of UUID is strongly recommended. For CIMXML data files in RDF syntax conforming to IEC 61970-552, the mRID is mapped to rdf:ID or rdf:about attributes that identify CIM object elements.
		--- The identification of a resource associated with a TimeSeries. --- Location of this power system resource. --- The geometry for a given

		location. X axis position.
registeredResource.location.geometry.positionPoint.xPosition (MBIE)	0..1 String	--- The identification of a resource associated with a TimeSeries. --- Location of this power system resource. --- The geometry for a given location. Y axis position.
registeredResource.location.geometry.positionPoint.yPosition (MBIE)	0..1 String	--- The identification of a resource associated with a TimeSeries. --- Location of this power system resource. --- The geometry for a given location. (if applicable) Z axis position.
registeredResource.location.geometry.positionPoint.zPosition (MBIE)	0..1 String	--- The identification of a resource associated with a TimeSeries. --- Location of this power system resource. --- The geometry for a given location. A Uniform Resource Name (URN) for the coordinate reference system (crs) used to define 'Location.PositionPoints'. An example would be the European Petroleum Survey Group (EPSG) code for a coordinate reference system, defined in URN under the Open Geospatial Consortium (OGC) namespace as: urn:ogc:def:uom:EPSG::XXXX, where XXXX is an EPSG code (a full list of codes can be found at the EPSG Registry web site http://www.epsg-registry.org/). To define the coordinate system as being WGS84 (latitude, longitude) using an EPSG OGC, this attribute would be urn:ogc:def:uom:EPSG::4236. A profile should limit this code to a set of allowed URNs agreed to by all sending and receiving parties.
registeredResource.location.geometry.coordinateSystem.crsUrn (MBIE)	0..1 String	--- The identification of a resource associated with a TimeSeries. --- Location of this power system resource. --- The geometry for a given location. --- The coordinate system used for a given geometry. An indication of the fuel used for the energy production, or part of the energy production, that is potentially fed into the

registeredResource.fuel.fuel (MBIE)	0..1 Fuel_String	grid at the related object. --- The identification of a resource associated with a TimeSeries. The motivation of an act in coded form.
reason.code (MBIE)	0..1 ReasonCode_String	--- The reason information associated with a TimeSeries providing motivation information. The textual explanation corresponding to the reason code.
reason.text (MBIE)	0..1 ReasonText_String	--- The reason information associated with a TimeSeries providing motivation information. The identification of the formal code for a measurement unit (UN/ECE Recommendation 20). --- The unit of measure associated with the quantities in a TimeSeries.
energy_Measurement_Unit.name (MBIE)	0..1 MeasurementUnitKind_String	The identification of the formal code for a measurement unit (UN/ECE Recommendation 20). --- The unit of measure associated with the quantities in a TimeSeries.
energyQuality_Measurement_Unit.name (MBIE)	0..1 MeasurementUnitKind_String	A unique identification of the measurement point. In the ESMP context, the "model authority" is defined as an authorized issuing office that provides an agreed identification coding scheme for market participant, domain, measurement point, resources (generator, lines, substations, etc.) identification. Master resource identifier issued by a model authority. The mRID is globally unique within an exchange context. Global uniqueness is easily achieved by using a UUID for the mRID. It is strongly recommended to do this. For CIMXML data files in RDF syntax, the mRID is mapped to rdf:ID or rdf:about attributes that identify CIM object elements. --- The identification of a measurement point associated with a TimeSeries.
accountingPoint.mRID (MBIE)	0..1 MeasurementPointID_String	The name is any free human readable and possibly non unique text naming the object. --- The identification of a measurement point associated with a TimeSeries. (if applicable) Reference to geographical information source, often external to the utility.
accountingPoint.name (MBIE)	0..1 String	

accountingPoint.usagePointLocation.geoInfoReference (MBIE)	0..1 String	--- The identification of a measurement point associated with a TimeSeries. --- Location of this usage point. The unique identification of the meter reading.
accountingPoint.meterReadings.mRID (MBIE)	0..1 ResourceID_String	In the ESMP context, the "model authority" is defined as a party (originator of the exchange) that provides an identification in the context of a business exchange. Master resource identifier issued by a model authority. The mRID is unique within an exchange context. Global uniqueness is easily achieved by using a UUID, as specified in RFC 4122, for the mRID. The use of UUID is strongly recommended. For CIMXML data files in RDF syntax conforming to IEC 61970-552 Edition 1, the mRID is mapped to rdf:ID or rdf:about attributes that identify CIM object elements.
accountingPoint.meterReadings.name (MBIE)	0..1 String	--- The identification of a measurement point associated with a TimeSeries. --- All meter readings obtained from this usage point. The name is any free human readable and possibly non unique text naming the object.
accountingPoint.meterReadings.readings.mRID (MBIE)	0..1 ResourceID_String	--- The identification of a measurement point associated with a TimeSeries. --- All meter readings obtained from this usage point. The identification of the reading. Master resource identifier issued by a model authority. The mRID is unique within an exchange context. Global uniqueness is easily achieved by using a UUID, as specified in RFC 4122, for the mRID. The use of UUID is strongly recommended. For CIMXML data files in RDF syntax conforming to IEC 61970-552 Edition 1, the mRID is mapped to rdf:ID or rdf:about attributes that identify CIM object elements.
		--- The identification of a measurement point associated with a TimeSeries. --- All meter readings obtained from this usage point. --- All reading values contained within this meter reading. Salient attribute of the reading

		data aggregated from individual endpoints. This is mainly used to define a mathematical operation carried out over 'macroPeriod', but may also be used to describe an attribute of the data when the 'macroPeriod' is not defined.
accountingPoint.meterReadings.readings.readingType.aggregate (MBIE)	0..1 AggregateKind	
		--- The identification of a measurement point associated with a TimeSeries. --- All meter readings obtained from this usage point. --- All reading values contained within this meter reading. --- Type information for this reading value. Accumulation behaviour of a reading over time, usually 'measuringPeriod', to be used with individual endpoints (as opposed to 'macroPeriod' and 'aggregate' that are used to describe aggregations of data from individual endpoints).
accountingPoint.meterReadings.readings.readingType.accumulation (MBIE)	0..1 AccumulationKind	
		--- The identification of a measurement point associated with a TimeSeries. --- All meter readings obtained from this usage point. --- All reading values contained within this meter reading. --- Type information for this reading value. Commodity being measured.
accountingPoint.meterReadings.readings.readingType.commodity (MBIE)	0..1 CommodityKind	
		--- The identification of a measurement point associated with a TimeSeries. --- All meter readings obtained from this usage point. --- All reading values contained within this meter reading. --- Type information for this reading value.

(MBIE) VHD_MarketDocument

An electronic document containing the information necessary to satisfy the requirements of a given business process.

!# shows all attributes of VHD_MarketDocument.

Table !# - Attributes of CEEDS_ValidatedHistoricalData Document assembly model::VHD_MarketDocument

Attribute name	mult.	Attribute type	Description
mRID (MBIE)	1..1	ID_String	The unique identification of the document being exchanged within a business process flow. In the ESMP context, the "model authority" is defined as a party (originator of the exchange) that provides an identification in the context of a business exchange such as document identification, ... Master resource identifier issued by a model authority. The mRID is globally unique within an exchange context.

			Global uniqueness is easily achieved by using a UUID for the mRID. It is strongly recommended to do this. For CIMXML data files in RDF syntax, the mRID is mapped to rdf:ID or rdf:about attributes that identify CIM object elements.
createdDateTime (MBIE)	1..1	ESMP_DateTime	The date and time of the creation of the document.
description (MBIE)	0..1	String	The description is a free human readable text describing or naming the object. It may be non unique and may not correlate to a naming hierarchy.
revisionNumber (MBIE)	0..1	ESMPVersion_String	The identification of the version that distinguishes one evolution of a document from another.
type (MBIE)	0..1	MessageKind_String	The coded type of a document. The document type describes the principal characteristic of the document.
process.processType (MBIE)	1..1	ProcessKind_String	The identification of the nature of process that the document addresses. --- The Process associated with an electronic document header that is valid for the whole document.
receiver_MarketParticipant.mRID (MBIE)	1..1	PartyID_String	The identification of a party in the energy market. In the ESMP context, the "model authority" is defined as an authorized issuing office that provides an agreed identification coding scheme for market participant, domain, measurement point, resources (generator, lines, substations, etc.) identification. Master resource identifier issued by a model authority. The mRID is globally unique within an exchange context. Global uniqueness is easily achieved by using a UUID for the mRID. It is strongly recommended to do this. For CIMXML data files in RDF syntax, the mRID is mapped to rdf:ID or rdf:about attributes that identify CIM object elements.
receiver_MarketParticipant.name (MBIE)	0..1	String	The name is any free human readable and possibly non unique text naming the object.
receiver_MarketParticipant.marketRole.type (MBIE)	1..1	MarketRoleKind_String	The identification of the role played by a market player.
sender_MarketParticipant.mRID (MBIE)	1..1	PartyID_String	The identification of a party in the energy market. In the ESMP context, the "model authority" is defined as an authorized issuing office that provides an agreed identification coding scheme for market participant, domain, measurement point, resources (generator, lines, substations, etc.) identification. Master resource identifier issued by a model authority. The mRID is globally unique within an exchange context. Global uniqueness is easily achieved by using a UUID for the mRID. It is strongly recommended to do this. For CIMXML data files in RDF syntax, the mRID is mapped to rdf:ID or rdf:about attributes that identify CIM object elements.
sender_MarketParticipant.name (MBIE)	0..1	String	The name is any free human readable and possibly non unique text naming the object.
sender_MarketParticipant.marketRole.type (MBIE)	1..1	MarketRoleKind_String	The identification of the role played by a market player.

Global Used Datatypes

Datatypes

MeasurementPointID_String

The coded identification of a domain covering a number of related objects, such as metering point, accounting point, etc.

In the ESMP context, it is an authorized issuing office that provides an agreed identification coding scheme for measurement point identification.

!# shows all attributes of MeasurementPointID_String.

Table !# - Attributes of ESMPDataTypes::MeasurementPointID_String.

Attribute name	mult.	Attribute type	Description
codingScheme (CIMDatatype)	1..1	CodingSchemeTypeList	DomainQualification.
value (CIMDatatype)	1..1	String	Main Core value Space.

DirectionKind_String

The coded identification of the direction of energy flow.

!# shows all attributes of DirectionKind_String.

Table !# - Attributes of ESMPDataTypes::DirectionKind_String.

Attribute name	mult.	Attribute type	Description
value (CIMDatatype)	1..1	DirectionTypeList	Main Core value Space.

Fuel_String

The identification of fuel used for the energy production.

!# shows all attributes of Fuel_String.

Table !# - Attributes of ESMPDataTypes::Fuel_String.

Attribute name	mult.	Attribute type	Description
value (CIMDatatype)	1..1	FuelTypeList	Main Core value Space.

PartyID_String

The identification of an actor in the energy market.

In the ESMP context, it is an authorized issuing office that provides an agreed identification coding scheme for market participant identification.

!# shows all attributes of PartyID_String.

Table !# - Attributes of ESMPDataTypes::PartyID_String.

Attribute name	mult.	Attribute type	Description
codingScheme (CIMDatatype)	1..1	CodingSchemeTypeList	DomainQualification.
value (CIMDatatype)	1..1	String	Main Core value Space.

MarketRoleKind_String

The identification of the role played by a party.

!# shows all attributes of MarketRoleKind_String.

Table !# - Attributes of ESMPDataTypes::MarketRoleKind_String.

Attribute name	mult.	Attribute type	Description
value (CIMDatatype)	1..1	RoleTypeList	Main Core value Space.

MeasurementUnitKind_String

The coded identification of a unit of measure that is applied to a quantity. The measurement units shall be in compliance with UN/ECE Recommendation 20.

!# shows all attributes of MeasurementUnitKind_String.

Table !# - Attributes of ESMPDataTypes::MeasurementUnitKind_String.

Attribute name	mult.	Attribute type	Description
value (CIMDatatype)	1..1	UnitOfMeasureTypeList	Main Core value Space.

ESMP_DateTime

In ESMP, the dateTime shall be expressed in UTC as YYYY-MM-DDThh:mm:ssZ.

!# shows all attributes of ESMP_DateTime.

Table !# - Attributes of ESMPDataTypes::ESMP_DateTime.

Attribute name	mult.	Attribute type	Description
value (CIMDatatype)	1..1	DateTime	Main Core value Space.

ResourceID_String

The identification of a resource object in the energy market.

In the ESMP context, it is an authorized issuing office that provides an agreed identification coding scheme for resources (generator, lines, substations, etc.) identification.

!# shows all attributes of ResourceID_String.

Table !# - Attributes of ESMPDataTypes::ResourceID_String.

Attribute name	mult.	Attribute type	Description
codingScheme (CIMDatatype)	1..1	CodingSchemeTypeList	DomainQualification.
value (CIMDatatype)	1..1	String	Main Core value Space.

PsrType_String

The coded type of a power system resource.

!# shows all attributes of PsrType_String.

Table !# - Attributes of ESMPDataTypes::PsrType_String.

Attribute name	mult.	Attribute type	Description
value (CIMDatatype)	1..1	AssetTypeList	Main Core value Space.

Position_Integer

An integer value, this value is used as a sequential value representing the relative position of an entity within a space such as a time interval.

!# shows all attributes of Position_Integer.

Table !# - Attributes of ESMPDataTypes::Position_Integer.

Attribute name	mult.	Attribute type	Description
value (CIMDatatype)	1..1	Integer	Main Core value Space.

ProcessKind_String

The coded identification of the nature of process.

!# shows all attributes of ProcessKind_String.

Table !# - Attributes of ESMPDataTypes::ProcessKind_String.

Attribute name	mult.	Attribute type	Description
value (CIMDatatype)	1..1	ProcessTypeList	Main Core value Space.

Quality_String

The coded identification of the quality of the information.

!# shows all attributes of Quality_String.

Table !# - Attributes of ESMPDataTypes::Quality_String.

Attribute name	mult.	Attribute type	Description
value (CIMDatatype)	1..1	QualityTypeList	Main Core value Space.

ReasonCode_String

The coded motivation of an act.

!# shows all attributes of ReasonCode_String.

Table !# - Attributes of ESMPDataTypes::ReasonCode_String.

Attribute name	mult.	Attribute type	Description
value (CIMDatatype)	1..1	ReasonCodeTypeList	Main Core value Space.

ReasonText_String

The textual explanation of an act as a string of characters.

!# shows all attributes of ReasonText_String.

Table !# - Attributes of ESMPDataTypes::ReasonText_String.

Attribute name	mult.	Attribute type	Description
value (CIMDatatype,IsBasedOn)	1..1	String	Main Core value Space.

YMDHM_DateTime

In ESMP, the date and time is expressed as "YYYY-MM-DDThh:mmZ", which conforms with ISO 8601 UTC time zone. This date and time is without the seconds.

!# shows all attributes of YMDHM_DateTime.

Table !# - Attributes of ESMPDataTypes::YMDHM_DateTime.

Attribute name	mult.	Attribute type	Description
value (CIMDatatype)	1..1	DateTime	The date and time as "YYYY-MM-DDThh:mmZ", which conforms with the ISO 8601 UTC time zone.

ID_String

A code to uniquely distinguish one occurrence of an entity from another.

In the ESMP context, the code is defined either by:

- an emitting company that provides an agreed identification unique within a business context such as capacity auction identification, market agreement identification, etc.

- a party (originator of the exchange) that provides a unique identification in the framework of a business exchange such as document identification, time series identification, bid identification, ...

!# shows all attributes of ID_String.

Table !# - Attributes of ESMPDataTypes::ID_String.

Attribute name	mult.	Attribute type	Description
value (CIMDatatype)	1..1	String	Main Core value Space.

ESMPVersion_String

In ESMP, the coded value is restricted to digits.

A code that distinguishes one evolution of an identified object from another. Information about a specific object may be sent several times, each transmission being identified by a different version number.

!# shows all attributes of ESMPVersion_String.

Table !# - Attributes of ESMPDataTypes::ESMPVersion_String.

Attribute name	mult.	Attribute type	Description
value (CIMDatatype)	1..1	String	Main Core value Space.

BusinessKind_String

The coded identification of the business type.

!# shows all attributes of BusinessKind_String.

Table !# - Attributes of ESMPDataTypes::BusinessKind_String.

Attribute name	mult.	Attribute type	Description
value (CIMDatatype)	1..1	BusinessTypeList	Main Core value Space.

EnergyProductKind_String

The identification of the nature of an energy product such as power, energy, reactive power, etc.

!# shows all attributes of EnergyProductKind_String.

Table !# - Attributes of ESMPDataTypes::EnergyProductKind_String.

Attribute name	mult.	Attribute type	Description
value (CIMDatatype)	1..1	EnergyProductTypeList	Main Core value Space.

MessageKind_String

The coded type of a document.

!# shows all attributes of MessageKind_String.

Table !# - Attributes of ESMPDataTypes::MessageKind_String.

Attribute name	mult.	Attribute type	Description

value (CIMDatatype) 1..1 MessageTypeList Main Core value Space.

Enumerations

CodingSchemeTypeList

Codification scheme used to identify the coding scheme used for the set of coded values to identify specific objects.

Literal	Description	Code
EIC	The coding scheme is the Energy Identification Coding Scheme (EIC), maintained by ENTSO-E.	A01
Coding scheme which are part of GS1 system	The coding schemes maintained by GS1.	A10
Andorra National coding scheme	The National coding scheme of the country in question.	NAD
Albania National coding scheme	The National coding scheme of the country in question.	NAL
Armenia National coding scheme	The National coding scheme of the country in question.	NAM
Austria National coding scheme	The National coding scheme of the country in question.	NAT
Azerbaijan National coding scheme	The National coding scheme of the country in question.	NAZ
Bosnia and Herzegovina National coding scheme	The National coding scheme of the country in question.	NBA
Belgium National coding scheme	The National coding scheme of the country in question.	NBE
Bulgaria National coding scheme	The National coding scheme of the country in question.	NBG
Switzerland National coding scheme	The National coding scheme of the country in question.	NCH
Serbia and Montenegro National coding scheme	The National coding scheme of the country in question.	NCS
Czech Republic National coding scheme	The National coding scheme of the country in question.	NCZ
Germany National coding scheme	The National coding scheme of the country in question.	NDE
Denmark National coding scheme	The National coding scheme of the country in question.	NDK
Estonia National coding scheme	The National coding scheme of the country in question.	NEE
Spain National coding scheme	The National coding scheme of the country in question.	NES
Finland National coding scheme	The National coding scheme of the country in question.	NFI
France National coding scheme	The National coding scheme of the country in question.	NFR
United Kingdom National coding scheme	The National coding scheme of the country in question.	NGB
Georgia National coding scheme	The National coding scheme of the country in question.	NGE
Gibraltar National coding scheme	The National coding scheme of the country in question.	NGI
Greece National coding scheme	The National coding scheme of the country in question.	NGR
Croatia National coding scheme	The National coding scheme of the country in question.	NHR
Hungary National coding scheme	The National coding scheme of the country in question.	NHU
Ireland National coding scheme	The National coding scheme of the country in question.	NIE
Italy National coding scheme	The National coding scheme of the country in question.	NIT
Kyrgyzstan National coding scheme	The National coding scheme of the country in question.	NKG
Kazakhstan National coding scheme	The National coding scheme of the country in question.	NKZ
Liechtenstein National coding scheme	The National coding scheme of the country in question.	NLI
Lithuania National coding scheme	The National coding scheme of the country in question.	NLT
Luxembourg National coding scheme	The National coding scheme of the country in question.	NLU
Latvia National coding scheme	The National coding scheme of the country in question.	NLV
Morocco National coding scheme	The National coding scheme of the country in question.	NMA
Moldavia National coding scheme	The National coding scheme of the country in question.	NMD
Macedonia National coding scheme	The National coding scheme of the country in question.	NMK
Netherlands National coding scheme	The National coding scheme of the country in question.	NNL
Nordic Regional coding scheme	The coding scheme of the Nordic region which covers Denmark, Finland, Norway and Sweden.	NNN
Norway National coding scheme	The National coding scheme of the country in question.	NNO

Poland National coding scheme	The National coding scheme of the country in question.	NPL
Portugal National coding scheme	The National coding scheme of the country in question.	NPT
Romania National coding scheme	The National coding scheme of the country in question.	NRO
Russian Federation National coding scheme	The National coding scheme of the country in question.	NRU
Sweden National coding scheme	The National coding scheme of the country in question.	NSE
Slovenia National coding scheme	The National coding scheme of the country in question.	NSI
Slovakia National coding scheme	The National coding scheme of the country in question.	NSK
Turkey National coding scheme	The National coding scheme of the country in question.	NTR
Ukraine National coding scheme	The National coding scheme of the country in question.	NUA
CGM	The coding scheme used for Common Grid Model Exchange Standard (CGMES).	A02
Cyprus National coding scheme	The National coding scheme of the country in question.	NCY
ISO	The coding scheme for the preceding attribute is a code maintained by International Organization for Standardization (ISO), such as â€œISO 3166 *English country names and code elements*â€ .	A03

DirectionTypeList

The coded identification of the direction of energy flow.

Literal	Description	Code
UP	Up signifies that the available power can be used by the Purchasing area to increase energy.	A01
DOWN	Down signifies that the available power can be used by the Purchasing area to decrease energy.	A02
UP and DOWN	Up and Down signifies that the UP and Down values are equal.	A03
Stable	The direction at a given instant in time is considered to be stable.	A04

FuelTypeList

The identification of the type of fuel.

Literal	Description	Code
Unspecified	Fuel that cannot be associated with any of available fuel codes.	A01
Renewable solid unspecified	Fuel produced in a solid form from renewable resources that cannot be associated with any of available renewable solid fuel codes.	A02
Renewable solid municipal waste	Solid waste of biological material produced by households, hospitals and the tertiary sector (in general all waste that resembles household waste).	A03
Renewable solid industrial and commercial waste	Solid waste generated by businesses, production units, bureaus and offices.	A04
Renewable solid wood	Solid wood energy resource.	A05
Renewable solid animal fats	Solid animal fats energy resource.	A06
Renewable solid biomass from agriculture	Solid by-products and residues from agriculture.	A07
Renewable liquid unspecified	Fuel produced in form of liquid from renewable resources that cannot be associated with any of available renewable liquid fuel codes.	A08
Renewable liquid municipal biodegradable waste	Liquid municipal waste capable of undergoing anaerobic or aerobic decomposition.	A09
Renewable liquid black liquor	Alkaline-spent liquor obtained from the digesters during the production of sulphate or soda pulp.	A10
Renewable liquid pure plant oil	Oil produced from plants, mainly from rapeseed and sunflower crops.	A11
Renewable liquid waste plant oil	Oil produced from plants that is no longer usable for its originally intended purpose.	A12
Renewable liquid refined vegetable oil	Oil produced from plants and transformed into fuel by refining process (hydrocracking or hydrogenation).	A13
Renewable gaseous unspecified	Fuel produced in form of gas from renewable resources that cannot be associated with any of available renewable gaseous fuel codes.	A14
Renewable gaseous landfill gas	Gas produced by digestion of land filled waste.	A15
Renewable gaseous sewage gas	Gas produced from the anaerobic fermentation of sewage sludge.	A16
Renewable gaseous agricultural gas	Gas produced from agricultural processes.	A17
Renewable gaseous gas from organic waste digestion	Gas produced by anaerobic digestion process from organic waste.	A18
Renewable gaseous process gas	Gas produced by an industrial process as a consequence.	A19
Renewable gaseous other biogenic	Gas produced by form of biogenic resource transformation that cannot be associated with any of available renewable gaseous fuel codes that use biogenic energy resource.	A20

sources		
Renewable heating and cooling solar	Sunlight energy source.	A21
Renewable heating and cooling geothermal	Geothermal energy source.	A22
Renewable heating and cooling aerothermal	Thermal energy obtained from air.	A23
Renewable heating and cooling hydrothermal	Thermal energy obtained from large body of water.	A24
Renewable heating and cooling process heat	Thermal energy obtained from process heat.	A25
Renewable mechanical unspecified	Energy from renewable resource transformed into mechanical energy that cannot be associated with any of available renewable mechanical fuel codes.	A26
Renewable mechanical wind	Kinetic energy of wind transformed into mechanical energy.	A27
Renewable mechanical hydro and marine	Falling or flowing water energy or energy derived from tidal movement, wave motion or ocean current.	A28
Fossil unspecified	Unspecified fossil energy source.	A29
Fossil solid unspecified	Unspecified fossil solid energy source.	A30
Fossil solid hard coal	Hard coal solid fossil energy source. This includes Anthracite, Bituminous coal, Coking coal, Coke-oven coke and Lignite coke.	A31
Fossil solid brown coal	Brown coal solid fossil energy source. This includes Sub-bituminous coal, Lignite, Brown coal briquette and Peat briquette.	A32
Fossil solid peat	Solid fossil peat energy source.	A33
Fossil solid municipal waste	Solid fossil municipal waste energy source.	A34
Fossil solid industrial and commercial waste	Solid fossil industrial and commercial waste energy source.	A35
Fossil liquid unspecified	Unspecified fossil liquid energy source.	A36
Fossil liquid crude oil	Liquid crude oil fossil energy source. This includes shale oil or other types.	A37
Fossil liquid natural gas liquids (NGL)	Liquid natural gas liquids (NGL) fossil energy source.	A38
Fossil liquid petroleum products	Liquid petroleum products energy source. This includes Ethane,Naphtha, Aviation gasoline , Motor gasoline, Aviation turbine fuel,Other kerosene,Gas/diesel oil , Fuel oil, low sulphur ,Fuel oil, high sulphur ,Liquid Petroleum Gas , Orimulsion , Bitumen ,Lubricants ,Petroleum coke , Refinery Feedstock.	A39
Fossil gaseous unspecified	Unspecified fossil gaseous energy source.	A40
Fossil gaseous natural gas	Fossil gaseous natural gas energy source.	A41
Fossil gaseous coal-derived gas	Fossil gaseous coal-derived gas energy source. This includes Blast furnace gas,Coke-oven gas or other types.	A42
Fossil gaseous petroleum products	Fossil gaseous petroleum products energy source. This includes Propane, Butane, Refinery gas, Chemical waste gas or other types.	A43
Fossil gaseous municipal gas plant	Fossil gaseous municipal gas plant energy source.	A44
Fossil gaseous process gas	Fossil gaseous process gas energy source. This includes Carbon monoxide, Methane , Hydrogen (fossil sourced) ,Phosphor gas ,Oxy gas and other types.	A45
Fossil heat unspecified	Unspecified fossil heat energy source.	A46
Fossil heat process heat	Fossil process heat energy source.	A47
Nuclear solid radioactive fuel	Solid Nuclear radioactive energy source. This includes UOX, AGR, MOX or other types.	A48
Gas synthesis unspecified	Unspecified gas synthesis energy source.	A49
Gas synthesis furnace gas	Gas synthesis from furnace gas energy source.	A50
Waste heat and cold unspecified	Unspecified Waste heat and cold energy source.	A51
Waste heat and cold By-product in industrial installation	Waste heat and cold from by-product in industrial installation energy source.	A52
Waste heat and cold By-product in power generation	Waste heat and cold from by-product in power generation energy source.	A53
Waste heat and cold By-product in tertiary sector	Waste heat and cold from by-product in tertiary sector energy source.	A54
Hydrogen	Hydrogen energy source.	A55

RoleTypeList

Identification of the role played by a party.

Literal	Description	Code
Trade responsible party Consumption	Refer to role model definitions in the ENTSO-E Harmonised Role Model Document.	A01

responsible party	Refer to role model definitions in the ENTSO-E Harmonised Role Model Document.	A02
Combined power exchange (not to be used)	This role is no longer in the ENTSO-E Harmonised Role Model Document.	A03
System operator	Refer to role model definitions in the ENTSO-E Harmonised Role Model Document.	A04
Imbalance settlement responsible	Refer to role model definitions in the ENTSO-E Harmonised Role Model Document.	A05
Production responsible party	Refer to role model definitions in the ENTSO-E Harmonised Role Model Document.	A06
Transmission capacity allocator	Refer to role model definitions in the ENTSO-E Harmonised Role Model Document.	A07
Balance responsible party	Refer to role model definitions in the ENTSO-E Harmonised Role Model Document.	A08
Metered data aggregator	Refer to role model definitions in the ENTSO-E Harmonised Role Model Document.	A09
Billing agent	Refer to role model definitions in the ENTSO-E Harmonised Role Model Document.	A10
Market operator	Refer to role model definitions in the ENTSO-E Harmonised Role Model Document.	A11
Energy supplier	Refer to role model definitions in the ENTSO-E Harmonised Role Model Document.	A12
Consumer	Refer to role model definitions in the ENTSO-E Harmonised Role Model Document.	A13
Control area operator	Refer to role model definitions in the ENTSO-E Harmonised Role Model Document.	A14
Control block operator	Refer to role model definitions in the ENTSO-E Harmonised Role Model Document.	A15
Coordination centre operator	Refer to role model definitions in the ENTSO-E Harmonised Role Model Document.	A16
Grid access provider	Refer to role model definitions in the ENTSO-E Harmonised Role Model Document.	A17
Grid operator	Refer to role model definitions in the ENTSO-E Harmonised Role Model Document.	A18
Meter administrator	Refer to role model definitions in the ENTSO-E Harmonised Role Model Document.	A19
Party connected to grid	Refer to role model definitions in the ENTSO-E Harmonised Role Model Document.	A20
Producer	Refer to role model definitions in the ENTSO-E Harmonised Role Model Document.	A21
Profile maintenance party	Refer to role model definitions in the ENTSO-E Harmonised Role Model Document.	A22
Meter operator	Refer to role model definitions in the ENTSO-E Harmonised Role Model Document.	A23
Metered data collector	Refer to role model definitions in the ENTSO-E Harmonised Role Model Document.	A24
Metered data responsible	Refer to role model definitions in the ENTSO-E Harmonised Role Model Document.	A25
Metering point administrator	Refer to role model definitions in the ENTSO-E Harmonised Role Model Document.	A26
Resource Provider	Refer to role model definitions in the ENTSO-E Harmonised Role Model Document.	A27
Scheduling coordinator	Refer to role model definitions in the ENTSO-E Harmonised Role Model Document.	A28
Capacity Trader	Refer to role model definitions in the ENTSO-E Harmonised Role Model Document.	A29
Interconnection Trade Responsible	Refer to role model definitions in the ENTSO-E Harmonised Role Model Document.	A30
Nomination Validator	Refer to role model definitions in the ENTSO-E Harmonised Role Model Document.	A31
Market information aggregator	Refer to role model definitions in the ENTSO-E Harmonised Role Model Document. A party that collects information from different sources and assembles it to provide a summary of the market.	A32
Information receiver	Refer to role model definitions in the ENTSO-E Harmonised Role Model Document. A party, not necessarily a market participant, which receives information about the market.	A33
Reserve Allocator	Refer to role model definitions in the ENTSO-E Harmonised Role Model Document. A party that informs the market of reserve requirements, receives tenders against the requirements and in compliance with the prequalification criteria, determines what tenders meet requirements and assigns tenders.	A34
MOL Responsible	Refer to role model definitions in the ENTSO-E Harmonised Role Model Document. A party that Informs the market of reserve requirements, receives tenders against the requirements and in compliance with the prequalification criteria, determines what tenders meet requirements and assigns tenders.	A35
Capacity Coordinator	A party, acting on behalf of the SOs involved, responsible for establishing a coordinated Offered Capacity and/or NTC and/or ATC between several Scheduling Areas.	A36
Reconciliation Accountable	Refer to role model definitions in the ENTSO-E Harmonised Role Model Document. A party that is financially accountable for the reconciled volume of energy products for a profiled local metering point.	A37
Reconciliation Responsible	A party that is responsible for reconciling, within a metering grid area, the volumes used in the imbalance settlement process for profiled metering points and the actual metered quantities.	A38
Data provider	A party that is responsible for providing information to a central authority.	A39
Local Issuing Office (LIO)	A party that is responsible for operating a Local Issuing Office (LIO).	A40
Central Issuing Office (CIO)	A party that is responsible for operating a Central Issuing Office (CIO).	A41
EIC Participant	A party that participates in the EIC environment.	A42
Weather analyser	A party that analyses the current and forecast weather situation and establishes a prognosis of its impact on the renewable energy environment as well as the overall load.	A43
Regional Security Coordinator (RSC)	The RSC as defined in the System Operation guideline.	A44
Energy Service	A party offering energy-related services to the Party Connected to Grid, but not directly active in the energy value chain or the physical infrastructure itself. The ESCO may provide insight services as well as energy management	A45

Company (ESCO)	services.	
Balancing Service Provider	A party with reserve-providing units or reserve-providing groups able to provide balancing services to LFC Operators.	A46
Energy trader	A party that is selling or buying energy.	A47
LFC Operator	A party that is responsible for the Load Frequency Control of its LFC Area or LFC Block.	A48
Transmission System Operator (TSO)	The Transmission System Operator (TSO) is responsible for the transport of electricity on the extra high-voltage and high-voltage interconnected system. This is a market participant and not a role in Harmonised Role Model.	A49
Distribution System Operator (DSO)	Distribution System Operator (DSO) is responsible for transport of electricity on high-voltage (optionally), medium-voltage and low-voltage distribution systems. This is a market participant and not a role in Harmonised Role Model.	A50
Resource Capacity Mechanism Operator	Resource capacity mechanism operator is the party responsible to operate the resource capacity mechanism in a member state. It can either be the TSO or an independent party.	A51
Resource aggregator	A party that aggregates resources for usage by a service provider for energy market services.	A52
Cost sharing calculator	A party responsible for the mapping of the costs between areas and parties.	A53
Settlement responsible	A party responsible for settling the costs.	A54
Coordinated Capacity Calculator	Coordinated Capacity Calculator is the entity or entities with the task of calculating transmission capacity, at regional level or above.	A55
Flexibility service provider	A party that offers flexibility services based on acquired (aggregated) resources.	A56
Data access provider	A party responsible for facilitating access to data.	A57
Metered data administrator	A party responsible for storing and distributing validated measured data.	A58
Permission administrator	A party responsible for administrating a register of data access permissions for a set of metering points, making this information available to final customers and eligible parties in the sector, on request.	A59

UnitOfMeasureTypeList

(synonym MeasurementUnit) The unit of measure that is applied to a quantity. The measurement units shall be in compliance with UN/ECE Recommendation 20.

Literal	Description	Code
ampere	The unit of electrical current in the International system of Units (SI) equivalent to one Coulomb per second.	AMP
megavolt-ampere	MVA unit as per UN/CEFACT recommendation 20.	MVA
One	A unit for dimensionless quantities, also called quantities of dimension one.	C62
cubic hectometres	A unit of volume equal to one million cubic metres.	HMQ
kilovolt	kV unit as per UN/CEFACT recommendation 20.	KVT
kilometre	km unit as per UN/CEFACT recommendation 20.	KMT
kilovolt ampere reactive	A unit of electrical reactive power represented by a current of one thousand amperes flowing due to a potential difference of one thousand volts where the sine of the phase angle between them is 1. The unity power factor is expressed in thousands of a volt ampere reactive.	KVR
gigawatt hour	GWh unit as per UN/CEFACT recommendation 20.	GWH
kilowatt hour	A total amount of electrical energy transferred or consumed in one hour.	KWH
kilowatt	A unit of bulk power flow, which can be defined as the rate of energy transfer /consumption when a current of 1000 amperes flows due to a potential of 1000 volts at unity power factor expressed in thousands of a watt.	KWT
megavolt ampere reactive hours	Total amount of reactive power across a power system.	MAH
megavolt ampere reactive	A unit of electrical reactive power represented by a current of one thousand amperes flowing due to a potential difference of one thousand volts where the sine of the phase angle between them is 1.	MAR
megawatt	A unit of bulk power flow, which can be defined as the rate of energy transfer /consumption when a current of 1000 amperes flows due to a potential of 1000 volts at unity power factor expressed in millions of a watt.	MAW
cubic metres per second	The volume flow rate of cubic metre per second.	MQS
cubic metre	A Cubic metre.	MTQ
metre	The length of a metre.	MTR
megawatt hours	The total amount of bulk energy transferred or consumed.	MWH
percent	A unit of proportion equal to 0.01.	P1
gigawatt	GW unit as per UN/CEFACT recommendation 20.	A90
watt	The watt is the International System of Units (SI) standard unit of power (energy per unit time), the equivalent of one joule per second.	WTT
meter per second	A unit of measurement of the speed expressed in m/s.	MTS
degree (unit of angle)	A unit of measurement of angles expressed in a 0 to 360 degree gradient.	DD
watt per square meter	A unit of measurement of the density of heat flow rate expressed in watt per square meter.	D54
Celsius	A unit of measurement of temperature expressed in degree Celsius.	CEL
OKTA unit	A unit of measurement of the cloudiness expressed in OKTA or OCTA, i.e. A unit of count defining the number of eighth-	A59

	parts as a measure of the celestial dome cloud coverage.	
hectopascal	A unit of measurement of the pressure expressed in hectopascal.	A97
millimeter	A unit of measurement of length expressed in millimeter.	MMT
K (Kelvin)	Temperature unit refer ISO 80000-5 (Quantities and units, Part 5: Thermodynamics).	KEL
Hertz	HTZ unit as per UN/CEFACT recommendation 20.	HTZ
Megawatt per Hertz	A unit of energy expressed as the load change in million watts that will cause a frequency shift of one hertz.	E08
millihertz	A unit of frequency equal to 0.001 cycle per second.	MTZ
minute	A period of time equal to sixty seconds.	MIN
second	A period of time equal to one second.	SEC

AssetTypeList

The identification of the type of asset.

Literal	Description	Code
Tie line	A high voltage line used for cross border energy interconnections.	A01
Line	A specific electric line within a country.	A02
Resource Object	A resource that can either produce or consume energy.	A03
Generation	A resource that can produce energy.	A04
Load	A resource that can consume energy.	A05
Phase Shift Transformer	An electrical device for controlling the power flow through specific lines in a power transmission network.	A06
Circuit Breaker	An electrical switch designed to protect an electrical circuit from damage caused by overcurrent/overload or short circuit.	A07
Biomass	A resource using biomass for energy.	B01
Fossil Brown coal/Lignite	A resource using Fossil Brown coal/Lignite for energy.	B02
Fossil Coal-derived gas	A resource using Fossil Coal-derived gas for energy.	B03
Fossil Gas	A resource using Fossil Gas for energy.	B04
Fossil Hard coal	A resource using Fossil Hard coal for energy.	B05
Fossil Oil	A resource using Fossil Oil for energy.	B06
Fossil Oil shale	A resource using Fossil Oil shale for energy.	B07
Fossil Peat	A resource using Fossil Peat for energy.	B08
Geothermal	A resource using Geothermal for energy.	B09
Hydro-electric pure pumped storage head installation	Unit in which moving water energy is converted to electricity using flowing water to generate electricity with a large dam and reservoirs. Pure pumped storage plants store water in an upper reservoir with no natural inflows.	B10
Hydro Run-of-river head installation	Unit in which moving water energy is converted to electricity using flowing water to generate electricity in the absence of a large dam and reservoirs.	B11
Hydro-electric storage head installation	Unit in which moving water energy is converted to electricity using flowing water to generate electricity with a large dam and reservoirs.	B12
Marine unspecified	Unit in which marine energy is converted to electricity with equipment/devices not specified.	B13
Nuclear unspecified	A unit in which the heat source is a nuclear reactor of type that is not specified in other nuclear types.	B14
Other renewable	A resource using Other renewable for energy.	B15
Solar unspecified	Unit in which solar energy is converted to electricity with equipment/devices not specified.	B16
Waste	A resource using Waste for energy.	B17
Wind Offshore	Unit in which wind energy is converted to electricity using wind farms constructed in bodies of water, usually in the ocean.	B18
Wind Onshore	Unit in which wind energy is converted to electricity using wind farms constructed on land.	B19
Other unspecified	Other unspecified technology.	B20
AC Link	Overhead line or cable which is used to transmit electrical power via Alternative Current.	B21
DC Link	Overhead line or cable which is used to transmit electrical power via Direct Current.	B22
Substation	An assembly of equipment in an electric power system through which electric energy is passed for transmission, transformation, distribution or switching.	B23
Transformer	Electrical device that transfers energy from one voltage level to another voltage level.	B24
Busbar	A specific element within a substation to connect grid elements for energy distribution purposes.	A08
Capacitor	A transmission element designed to inject reactive power into the transmission network.	A09
Inductor	A transmission element designed to compensate reactive power in the transmission network.	A10
Power plant connection	All the network equipment that link the generating unit to the grid.	A11
FACTS	Flexible Alternating Current Transmission System	A12
Energy storage	A resource that stores energy. It could be gas, electricity, etc.	B25
Demand Side Response	A resource that change its electricity consumption patterns in response to a signal or incentive.	B26
Production unit	A production unit is a composition of one or several generation units.	A13

Dispatchable hydro resource	A resource referring to dispatchable hydro generation.	B27
Solar photovoltaic	Unit in which solar energy is converted to electricity using a technology based on the photoelectric effect.	B28
Solar concentration	Unit in which solar energy is converted to electricity using mirrors to concentrate the sun's energy to drive traditional steam turbines or engines.	B29
Wind unspecified	Unit in which wind energy is converted to electricity with equipment/devices not specified.	B30
Hydro-electric unspecified	Unit in which moving water energy is converted to electricity with equipment/devices not specified.	B31
Hydro-electric mixed pumped storage head installation	Unit in which moving water energy is converted to electricity using flowing water to generate electricity with a large dam and reservoirs. Mixed pumped storage plants use a combination of pumped storage and conventional hydroelectric plants with an upper reservoir that is replenished in part by natural inflows from a stream or river.	B32
Marine tidal	Unit in which marine energy from tides is converted to electricity.	B33
Marine wave	Unit in which marine energy from waves is converted to electricity.	B34
Marine currents	Unit in which marine energy from currents is converted to electricity.	B35
Marine pressure	Unit in which marine energy from pressure is converted to electricity.	B36
Thermal unspecified	Unit in which heat energy is converted to electricity with equipment/devices not specified in other thermal types.	B37
Thermal combined cycle gas turbine with heat recovery	Unit in which heat energy is converted to electricity called Combined Cycle Gas Turbine. The power is generated by the single or multiple gas turbine(s) in combination with the steam turbine(s). The unit might be equipped with waste heat recovery (e.g. to district heating network).	B38
Thermal steam turbine with back-pressure turbine (open cycle)	Unit in which heat energy is converted to electricity. The power is generated with the steam that is expanded in the back-pressure steam turbine with or without heat output (e.g. to district heating network).	B39
Thermal steam turbine with condensation turbine (closed cycle)	Unit in which heat energy is converted to electricity. The power is generated with the steam that is expanded in the condensation steam turbine with or without heat output (e.g. to district heating network).	B40
Thermal gas turbine with heat recovery	Unit in which heat energy is converted to electricity called Simple Cycle Gas Turbine. The power is generated by the gas turbine and the flue gas waste heat is recovered (e.g. to district heating network).	B41
Thermal internal combustion engine	An internal combustion engine is a heat engine in which the combustion of a fuel occurs with an oxidizer (usually air) in a combustion chamber that is an integral part of the working fluid flow circuit (e.g. reciprocating engine). The unit might be equipped with waste heat recovery (e.g. to district heating network).	B42
Thermal micro-turbine	Unit in which heat energy is converted to electricity called Simple Cycle Gas Turbine. The power is generated by the gas turbine (capacity less than 500kWe). The unit might be equipped with waste heat recovery (e.g. to district heating network).	B43
Thermal Stirling engine	A Stirling engine is a heat engine that is operated by the cyclic compression and expansion of air or other gas (the working fluid) at different temperatures, resulting in a net conversion of heat energy to mechanical work.	B44
Thermal fuel cell	A fuel cell is an electrochemical cell that converts the chemical energy of a fuel (e.g. hydrogen) and an oxidizing agent (e.g. oxygen) into electricity through a pair of redox reactions.	B45
Thermal steam engine	A steam engine is a heat engine that performs mechanical work using steam as its working fluid. The steam engine uses the force produced by steam pressure to push a piston back and forth inside a cylinder.	B46
Thermal organic Rankine cycle	The Organic Rankine Cycle (ORC) is named for its use of an organic, high molecular mass fluid with a liquid-vapor phase change, or boiling point, occurring at a lower temperature than the water-steam phase change. The fluid allows Rankine cycle heat recovery from lower temperature sources such as biomass combustion, industrial waste heat, geothermal heat, solar ponds etc. The low-temperature heat is converted into useful work, that can itself be converted into electricity.	B47
Thermal gas turbine without heat recovery	Unit in which heat energy is converted to electricity called Simple Cycle Gas Turbine. The power is generated by the gas turbine and there is no flue gas waste heat recovery.	B48
Nuclear heavy water reactor	A unit in which the heat source is a pressurized heavy-water reactor (PHWR) that is a nuclear reactor that uses heavy water (deuterium oxide D2O) as its coolant and neutron moderator.	B49
Nuclear light water reactor	A unit in which the heat source is a light-water reactor (LWR) that is a type of thermal-neutron reactor that uses normal water, as both its coolant and neutron moderator ? furthermore a solid form of fissile elements is used as fuel.	B50
Nuclear breeder	A unit in which the heat source is a nuclear reactor that generates more fissile material than it consumes.	B51
Nuclear graphite reactor	A unit in which the heat source is a graphite-moderated reactor that is a nuclear reactor that uses carbon as a neutron moderator, which allows natural uranium to be used as nuclear fuel.	B52
Internal tie line	An internal tie line is a line between two scheduling areas within the same bidding zone.	A14
Temporary energy storage	A resource that is temporarily connected to the grid and that may store energy when connected, such as an electric vehicle.	B53
Permanent energy storage	A resource that is permanently connected to the grid and that may store energy when connected, such as a pumped hydro.	B54
Electric vehicle battery	A resource using electric vehicle batteries, commercial and private. The reason for separating vehicle batteries and non-vehicle batteries is that the vehicle batteries not necessarily is connected to the charger.	B55
Heat pump specified	A heat pump is a device that uses work to transfer heat from a cool space to a warm space by transferring thermal energy using a refrigeration cycle	B56
Heat pump electrical	A heat pump is a device that uses electricity to transfer heat from a cool space to a warm space by transferring thermal energy using a refrigeration cycle.	B57
Heat pump	A heat pump is a device that uses absorption technology to transfer heat from a cool space to a warm space by	B58

absorption	transferring thermal energy using a refrigeration cycle.	
Auxiliary power unit	A technology that provides energy as a backup.	B59
Water electrolysis unspecified	Unspecified water electrolysis.	B60
Water electrolysis low temperature unspecified	Unspecified water electrolysis at low temperature as in an Alkaline or Proto-Exchange Membrane (PEM) fuel cell.	B61
Water electrolysis low temperature main product	Main product water electrolysis at low temperature as in an Alkaline or Proto-Exchange Membrane (PEM) fuel cell.	B62
Water electrolysis high temperature unspecified	Unspecified water electrolysis at high temperature as in a Solid Oxide Electrolysis Cell (SOEC) fuel cell.	B63
Steam methane reforming unspecified	Unspecified methane reforming.	B64
Steam methane reforming without CCS/CCU unspecified	Main product methane reforming without Carbon Capture and Sequestration (CCS)/Carbon Capture and Use (CCU).	B65
Steam methane reforming with CCS/CCU unspecified	Unspecified methane reforming with Carbon Capture and Sequestration (CCS)/Carbon Capture and Use (CCU).	B66
Steam methane reforming with CCS/CCU main product	Main product methane reforming with Carbon Capture and Sequestration (CCS)/Carbon Capture and Use (CCU).	B67
Partial oxidation unspecified	Unspecified partial oxidation.	B68
Autothermal reforming unspecified	Unspecified autothermal reforming.	B69
Methanol reforming unspecified	Unspecified methanol reforming.	B70
Ammonia reforming unspecified	Unspecified ammonia reforming.	B71
Ammonia gasification	Unspecified gasification.	B72
Chlor-alkali electrolysis unspecified	Unspecified alkali electrolysis.	B73
Chlor-alkali electrolysis by-product	Alkali electrolysis product.	B74
ACDC converter	ACDC converters are electrical circuits that transform alternating current (AC) into direct current (DC) and vice versa.	B75
Converter	Electrical device that converts current between AC and DC.	B76

ProcessTypeList

Indicates the nature of process that the document addresses.

Literal	Description	Code
Day ahead	The information provided concerns a day ahead process.	A01
Intra day incremental	The information provided concerns an intra day schedule.	A02
Inter-area transit	The information provided concerns an inter area transit schedule. The rules governing this process are market dependent	A03
System operation closure	The information provided concerns the closure of a given period of both scheduled and regulation information.	A04
Metered data aggregation	The information provided concerns the aggregation process of metered information.	A05
Imbalance settlement	The information provided concerns the imbalance settlement for a given period for a balance responsible party or parties.	A06
Capacity allocation	The information provided concerns the capacity allocation process.	A07
Central reconciliation	The process carried out to finalise the imbalance settlement based on actual metered values against provisional values from profiled metering points.	A08
Released capacity allocation	The process concerns the notification of capacity rights that are being released.	A09
Proposed capacity allocation	The process concerns the proposed capacity to be allocated for a given border.	A10
Agreed capacity allocation	The process concerns the capacity that has been agreed for allocation for a border.	A11
Long term	The process concerns scheduling all schedules except daily and intraday contracts.	A12
Post scheduling adjustment	The process concerns the adjustments made to previous schedules after the period of execution.	A13
Forecast	The data contained in the document are to be handled in short, medium, long term forecasting process.	A14
Capacity determination	The process of determining the capacity for use.	A15

Realised	The process for the treatment of realised data as opposed to forecast data.	A16
Schedule day	The process concerns the day ahead, intraday and eventually ex-post scheduling in a single document. The schedule will be transferred within the total position including historic information.	A17
Intraday total	This process concerns an intraday schedule which contains the accumulated day ahead and intraday current position.	A18
Intraday accumulated	This process concerns a single intraday schedule process where only intraday evolutions occur through version changes.	A19
SOMA process	System Operator meter alignment process.	A20
SOVM process	System Operator validated measurement process.	A21
RGCE accounting process	The document provides ENTSO-E Regional Group Continental Europe accounting process information.	A22
CCSR RGCE Settlement	The process concerns the control center settlement report for the whole of the ENTSO-E Regional Group Continental Europe.	A23
CBSR Settlement	The process concerns the control block settlement report.	A24
CASR Settlement	The process concerns the control area settlement report.	A25
Outage information	The process concerns TSO publication of outages on its power system.	A26
Reserve resource process	The process being described is for general reserve resources.	A27
Primary reserve process	The process being described is for primary reserves.	A28
Secondary reserve process	The process being described is for secondary reserves.	A29
Tertiary reserveprocess	The process being described is for tertiary reserves.	A30
Week ahead	The process being described is for the week ahead.	A31
Month ahead	The process being described is for the month ahead.	A32
Year ahead	The process being described is for the year ahead.	A33
Contracted	The process being described is for contracted information.	A34
Network information	The process being described is for network information.	A35
Creation	The process being described is for the creation of information.	A36
Modification	The process being described is for the modification of information.	A37
Deactivation process	The process being described is for deactivation of information.	A38
Synchronisation process	The process being described is for the synchronisation of information.	A39
Intraday process	The process being described is for intraday process.	A40
Redispatch process	The process being described is for redispatch activation.	A41
Activation history process	The process being described is for the provision of an activation history.	A42
Flow based domain constraint day-ahead	The information provided concerns the flow-based process in day ahead.	A43
Flow based domain constraint intraday	The information provided concerns the flow-based process in intraday.	A44
Two days ahead	Two days ahead.	A45
Replacement reserve	A process being described is for replacement reserves (RR) to restore or support the required level of frequency restoration reserves.	A46
Manual frequency restoration reserve	A process being described is for manual frequency restoration reserves (mFRR).	A47
Day-ahead capacity determination	The process run at the day-ahead timeframe to determine the capacity for use.	A48
Intraday capacity determination	The process run at the intraday timeframe to determine the capacity for use.	A49
Long term capacity determination	The process run at the long term timeframe to determine the capacity for use.	A50
Automatic frequency restoration reserve	A process being described is for automatic frequency restoration reserves (aFRR).	A51
Frequency containment reserve	A process being described is for frequency containment reserves (FCR).	A52
Common Grid Model (CGM) merging process	The process for merging individual grid models to form the common grid model.	A53
Coordinated security analysis	The process being described is used for coordinated security analysis.	A54
Exchange of master data	A process for exchanging master data.	A55
Frequency restoration reserve	The process being described is for general frequency restoration reserve.	A56
FSKAR settlement	The information provided concerns the Financial Settlement of K?f, ACE and ramping period settlement for a given period.	A57
mFRR capacity market	Processes related to the Reserve option market to assure that there are enough available reserves for the manual FRR market.	A58
Internal trade reporting	The process related to internal trade reporting.	A59
Scheduled activation mFRR	mFRR being subject to scheduled activation.	A60
Direct activation mFRR	mFRR being subject to direct activation.	A61
Registration	A process related to the registration and management of object information.	A62
Imbalance Netting	The process described is for imbalance netting.	A63
Criteria application for instantaneous frequency	The process describes criteria application for instantaneous frequency.	A64
Criteria application for frequency restoration	The process describes criteria application for frequency restoration.	A65
Cost sharing	The process describes the cost sharing process for costly remedial actions.	A66
Central selection aFRR	aFRR subject to central selection of bids for activation.	A67
Local selection aFRR	aFRR subject to local selection of bids for activation.	A68
Common grid model alignment	The process for aligning bidding zone net positions.	A69

Pan-European verification function	The process for verifying bidding zone net positions.	A70
Outage planning coordination	The process being described is used for outage planning coordination.	A71
Short term adequacy	The process being described is used for short term adequacy calculations.	A72
Coordinated capacity calculation	The process being described is used for coordinated capacity calculation.	A73
Access to metered data	Access to meter readings of electricity consumption from or electricity feed into the grid.	A74
Permission administration	Administration of permissions to access and/or use data.	A75

QualityTypeList

The quality of an object.

Literal	Description	Code
Adjusted	The contents of the object have been adjusted.	A01
Not available	The contents of the object are not available.	A02
Estimated	The contents of the object are estimated. The code is typically used when measured values are missing and an estimate is made based on historical data.	A03
As provided	The contents of the object are as provided.	A04
Incomplete	The contents of the object are calculated based on incomplete data.	A05
Calculated	The contents of the object are calculated. The code is typically used when a value is calculated based on several other known values.	A06
Temporary	The content of the object is temporary. The code is typically used when there is communication trouble to a meter and a more correct value is expected later.	A07

CommodityDirectionKind

CommodityDirectionKind is an enumeration that documents two key features for a specific DER operating mode: (a) the energy flow convention for a positive number as either production or consumption of service and (b) the sense of how the bid/offer amount is used as either a target or setpoint or as a relative change to a defined level.

Literal	Description	Code
production	The resource operates as a generation resource, where a positive amount corresponds to an injection of energy and/or grid support services to the grid. The amount of energy is an absolute level understood to be a production setpoint. Negative values are acceptable and represent an energy withdrawal setpoint. For energy storage resources, a positive amount represents a setpoint in the discharging state.	
consumption	The resource operates as a load resource, where a positive amount corresponds to a withdrawal of energy and/or support services from the grid. The amount of energy is an absolute level understood to be a consumption setpoint. Negative values are acceptable and represent an energy injection setpoint. For energy storage resources, a positive amount represents a setpoint in the charging state.	
consumptionIncrease	The resource operates as a variable load, where a positive amount corresponds to a relative load increase. The level is relative, generally as an upward offset to a reference level such as the current consumption or a calculated baseline consumption. Negative values are acceptable and represent a load reduction offset. For energy storage resources, a positive amount represents a faster rate of charge, a state change from discharging (or offline) to charge, or a slower rate of discharge.	
consumptionDecrease	The resource operates as a curtailable load, where a positive amount corresponds to a relative load reduction. The level is relative, generally as an downward offset to a reference level such as the current consumption or a calculated baseline consumption. Negative values are acceptable and represent a load increase offset. For energy storage resources, a positive amount represents a slower rate of charge, a state change from charging (or offline) to discharging, or a faster rate of discharge.	
productionIncrease	The resource operates primarily as a variable generation resource, where a positive amount corresponds to a relative generation increase.	
productionDecrease	The resource operates primarily as a variable generation resource, where a positive amount corresponds to a relative generation reduction.	

AggregateKind

Kind of aggregation for read / measured values from multiple end points.

Literal	Description	Code
none	Not applicable.	0
average	The value represents average.	2
excess	The value represents an amount over which a threshold was exceeded.	4
highThreshold	The value represents a programmed high threshold.	5
lowThreshold	The value represents a programmed low threshold.	7
maximum	The highest value observed.	8
minimum	The smallest value observed.	9
nominal	The nominal value.	11
normal	The normal value.	12
secondMaximum	The second highest value observed.	16

secondMinimum	The second smallest value observed.	17
thirdMaximum	The third highest value observed.	23
fourthMaximum	The fourth highest value observed.	24
fifthMaximum	The fifth highest value observed.	25
sum	The accumulated sum.	26

AccumulationKind

Kind of accumulation behaviour for read / measured values from individual end points.

Literal	Description	Code
none	Not applicable, or implied by the unit of measure.	0
bulkQuantity	A value from a register which represents the bulk quantity of a commodity. This quantity is computed as the integral of the commodity usage rate. This value is typically used as the basis for the dial reading at the meter, and as a result, will roll over upon reaching a maximum dial value. Note 1: With the metering system, the roll-over behaviour typically implies a roll-under behavior so that the value presented is always a positive value (e.g. unsigned integer or positive decimal.) However, when communicating data between enterprise applications a negative value might occur in a case such as net metering. Note 2: A 'bulkQuantity' refers primarily to the dial reading and not the consumption over a specific period of time.	1
continuousCumulative	The sum of the previous billing period values and the present period value. Note: 'continuousCumulative' is commonly used in conjunction with 'demand', and it would represent the cumulative sum of the previous billing period maximum demand values (as occurring with each demand reset) summed with the present period maximum demand value (which has yet to be reset.)	2
cumulative	The sum of the previous billing period values. Note: 'cumulative' is commonly used in conjunction with â€œdemand.â€ Each demand reset causes the maximum demand value for the present billing period (since the last demand reset) to accumulate as an accumulative total of all maximum demands. So instead of 'zeroing' the demand register, a demand reset has the effect of adding the present maximum demand to this accumulating total.	3
deltaData	The difference between the value at the end of the prescribed interval and the beginning of the interval. This is used for incremental interval data. Note: One common application would be for load profile data, another use might be to report the number of events within an interval (such as the number of equipment energisations within the specified period of time.)	4
indicating	As if a needle is swung out on the meter face to a value to indicate the current value. Note: An 'indicating' value is typically measured over hundreds of milliseconds or greater, or may imply a â€œpusherâ€ mechanism to capture a value. Compare this to 'instantaneous' which is measured over a shorter period of time.	6
summation	A form of accumulation which is selective with respect to time. Note : 'summation' could be considered a specialisation of 'bulkQuantity' as it selectively accumulates pulses over a timing pattern (while 'bulkQuantity' accumulates pulses all of the time).	9
timeDelay	A form of computation which introduces a time delay characteristic to the data value.	10
instantaneous	Typically measured over the fastest period of time allowed by the definition of the metric (usually milliseconds or tens of milliseconds.) Note: 'instantaneous' was moved to attribute #3 in Ed.2 of IEC 61968-9, from attribute #1 in Ed.1 of IEC 61968-9.	12
latchingQuantity	When this description is applied to a metered value, it implies that the value is a time-independent cumulative quantity much like a 'bulkQuantity', except that it latches upon the maximum value upon reaching that value. Any additional accumulation (positive or negative) is discarded until a reset occurs. Note: A 'latchingQuantity' may also occur in the downward direction â€“ upon reaching a minimum value. The terms 'maximum' or 'minimum' (for 'aggregate') will usually be included when this type of accumulation behaviour is present. When this description is applied to an encoded value (UOM= 'Code'), it implies that the value has one or more bits which are latching. The condition that caused the bit to be set may have long since evaporated. In either case, the timestamp that accompanies the value may not coincide with the moment the value was initially set. In both cases a system will need to perform an operation to clear the latched value.	13
boundedQuantity	A time-independent cumulative quantity much like a 'bulkQuantity' or a 'latchingQuantity', except that the accumulation stops at the maximum or minimum values. When the maximum is reached, any additional positive accumulation is discarded, but negative accumulation may be accepted (thus lowering the counter.) Likewise, when the negative bound is reached, any additional negative accumulation is discarded, but positive accumulation is accepted (thus increasing the counter.)	14

CommodityKind

Kind of commodity being measured.

Literal	Description	Code
H2	CommodityKind Hydrogen (H2)	26
none	Not Applicable	0
electricitySecondaryMetered	All types of metered quantities. This type of reading comes from the meter and represents a â€œsecondaryâ€ metered value.	1

electricityPrimaryMetered	It is possible for a meter to be outfitted with an external VT and/or CT. The meter might not be aware of these devices, and the display not compensate for their presence. Ultimately, when these scalars are applied, the value that represents the service value is called the â€œprimary meteredâ€ value. The â€œindexâ€ in sub-category 3 mirrors those of sub-category 0.	2
communication	A measurement of the communication infrastructure itself.	3
air		4
insulativeGas	(SF6 is found separately below.)	5
insulativeOil		6
naturalGas		7
propane		8
potableWater	Drinkable water	9
steam	Water in steam form, usually used for heating.	10
wasteWater	(Sewerage)	11
heatingFluid	This fluid is likely in liquid form. It is not necessarily water or water based. The warm fluid returns cooler than when it was sent. The heat conveyed may be metered.	12
coolingFluid	The cool fluid returns warmer than when it was sent. The heat conveyed may be metered.	13
nonpotableWater	Reclaimed water â€“ possibly used for irrigation but not sufficiently treated to be considered safe for drinking.	14
nox	Nitrous Oxides NOX	15
so2	Sulfur Dioxide SO2	16
ch4	Methane CH4	17
co2	Carbon Dioxide CO2	18
carbon		19
hch	Hexachlorocyclohexane HCH	20
pfc	Perfluorocarbons PFC	21
sf6	Sulfurhexafluoride SF6	22
tvLicence	Television	23
internet	Internet service	24
refuse	trash	25

ReasonCodeTypeList

The coded motivation of an act.

Literal	Description	Code
Errors not specifically identified	This code is used to identify errors that have not been specifically addressed in the Reason code list. It can be used at any level and refers to the level for which it has been identified.	999
Message fully accepted	The message has been fully accepted for application processing.	A01
Message fully rejected	No part of the message has been accepted for application processing, e.g. Global position incomplete.	A02
Message contains errors at the time series level	Part of the message contents, i.e. certain time series, has been accepted for application processing. It is necessary to look at the time series level to determine the time series that have been rejected. The time series is excluded from the global position.	A03
Time interval incorrect	The schedule time interval is not within the contractual agreement or the period does not agree with the schedule time interval.	A04
Sender without valid contract	The sender has no current valid contract with the TSO. The message consequently will be fully rejected.	A05
Schedule accepted	The schedule of the recipient as presented has been completely accepted.	A06
Schedule partially accepted	The schedule of the recipient as presented has been partially accepted. It is necessary to look at the time series level to determine the changes (time series rejected, modified, etc.).	A07
Schedule rejected	The schedule of the recipient as presented has been totally rejected. The cause could be the non presentation of a counter part for the involved trades.	A08
Time series not matching	Time series mismatches.	A09
Credit limit exceeded	The contractual credit limit has been exceeded and as such the message has been rejected.	A10
Time series fully rejected	The time series has been fully rejected. In the case of a confirmation report, this reason code is used in conjunction with either A26 or A30.	A20
Time series accepted with specific time interval errors	The time series has been accepted but some time interval quantities have been rectified or zeroed out.	A21
In party/Out party invalid	There is no contract for the parties indicated or the rules for cross border nominations are not being respected. The time series has been rejected.	A22
Area invalid	The area is unknown or not allowed. The time series has been rejected.	A23
A24 not applicable	This code is nolonger applicable.	A24
A25 not applicable	This code is nolonger applicable.	A25
Default time series applied	The time series has been rejected and replaced with a default time series profile. This reason code may not be used in conjunction with A30.	A26
Cross border capacity exceeded	The cross border capacity has been exceeded. The time series has been rejected or adjusted.	A27
Counterpart time series	This provides an indication that the time series has not got a counterpart time series. In the case of an Intermediate Confirmation Report this is advising the recipient that the time series may be rejected at nomination closure if the counterpart time series is not received.	A28

missing	In the case of a Final Confirmation Report this is informing the recipient that the time series has been rejected because the counterpart time series has not been forthcoming.	
Counterpart time series quantity differences	The time series has been rejected as it does not match that of the counterpart who is considered by market rules to be correct.	A29
Imposed Time series from nominated party's time series (party identified in reason text)	The nominated party's time series has replaced the current time series. This reason code may not be used in conjunction with A26.	A30
Resolution inconsistency	The resolution is not coherent with the time interval, or resolution not valid.	A41
Quantity inconsistency	The quantity is not coherent. For example a time period with the same version number but different quantities or an non permitted number of digits after the decimal point, etc.	A42
Quantity increased	The quantity has been increased in order to satisfy minimum constraints.	A43
Quantity decreased	The quantity has been decreased in order to satisfy congestion constraints.	A44
Default quantity applied	The default quantity has been applied as the current quantity does not satisfy contractual obligations.	A45
Quantities must not be signed values	The quantity proposed is illegal since signed values are only permitted in specific circumstances.	A46
A47 not applicable	This code is no longer applicable.	A47
Modification reason	In an intraday transmission, the reason for the modification is as follows (in the reason text).	A48
Position inconsistency	A position is missing or too many.	A49
Senders time series version conflict	There is an error in the sender time series version, i.e. it could be superior to the message version or is inconsistent with the existing data. The time series has been rejected.	A50
Message identification or version conflict	The message identification is already in the receiving system. Or a higher version already exists. Message rejected.	A51
Time series missing from new version of message	A time series is not contained in a new version of the message. Message rejected.	A52
Receiving party incorrect	The receiving party is incorrect. Message rejected.	A53
Global position not in balance	The message does not balance out to zero. Market rules might require that the message is rejected.	A54
Time series identification conflict	The identification of the time series is duplicated or incorrect. Time series will be rejected.	A55
Corresponding Time series not netted	All corresponding time series must be netted. Time series rejected.	A56
Deadline limit exceeded/Gate not open	The deadline for receiving schedule messages has passed. Message or time series rejected.	A57
One to one nomination inconsistency	There is a one to one nomination inconstancy with the in/out parties or areas. Time series rejected.	A58
Not compliant to local market rules	The level in which this is identified is not in compliance with local market rules. The level in question has been rejected.	A59
Inter-area transit schedule exceeds nominated schedule	The inter-area transit schedule exceeds the nominated schedule for the same time interval. The inter-area transit schedule is rejected.	A60
Currency invalid	The currency is not in compliance with ISO 4217.	A61
Invalid business type	The business type does not belong to the valid set of business types for the process in question.	A62
Time Series modified	The time series has been modified.	A63
Resource Object Invalid	The Resource Object defined in the document is not valid.	A64
Reserve object Technical limits exceeded	Reserve objects aggregated values are not within technical/prequalified limits	A65
Planned reserves do not correspond with contractual data	Planned reserves do not correspond with contractual data.	A66
Limit Data is not available	Limit Data is not available.	A67
Reserve Object not qualified for reserve type	Reserve Object is not qualified for reserve type.	A68
Mandatory attributes missing	Mandatory attributes missing.	A69
Curtailment	The capacity in question has been curtailed.	A70
Linked bid rejected due to associated bid unsuccessful	The bid in question has been rejected because an associated bid has been unsuccessful.	A71
Original bid divided to permit acceptance	The original bid quantity has been divided to enable it to be accepted.	A72
Bid accepted	The bid in question has been accepted.	A73
Auction Status	The information in the Reason Text provides auction status information.	A74
Right status information	The information in the Reason Text provides status information concerning the transmission rights in question.	A75
Agreement identification inconsistency	There is an inconsistency between the contract type and the agreement identification.	A76
Dependency matrix not respected	There is an inconsistency between the document contents and the dependency matrix.	A77
Sender identification and/or role invalid	The identification of the sender or the sender/role combination is invalid.	A78
Process type invalid	The process type does not figure in the list of valid process types for this document.	A79
Domain invalid	The domain does not figure in the list of valid domains for this document and process.	A80
Matching period invalid	The period is not within the expected limits.	A81
In/Out area inconsistent with domain	The in and out area does not figure within the domain specified.	A82
Disagree with matching results	The matching results provided are not consistent.	A83
Confirmation ignored due to higher version already	The report has been ignored since a higher version has been received.	A84

received		
Confirmation without adjustment (time series have been matched without change)	The report has been successfully matched without any changes.	A85
Confirmation with adjustment (time series have been modified)	The report has been matched but required adjustment.	A86
For action (only in intermediate confirmation - time series need mutual agreement and action)	The report in question is only for action in an intermediate stage.	A87
Time series matched	The time series has been successfully matched.	A88
Time series ignored (note: this can only apply to time series that are set to zero - see matching principles)	The time series has been ignored and not matched since it does not figure in a counterparty transmission. All are correctly equal to zero.	A89
Modification proposal (intermediate confirmation)	The document is a proposal for change before finalization.	A90
Expected document not received	The document that is expected has not been received within the expected timeframe.	A91
Not possible to send document on time, but estimated delivery time is provided	The document that is due cannot be transmitted within the required timeframe. An estimated time of transmission is provided.	A92
Not possible to send document on time, and furthermore no expected time of return to normal situation	The document that is due cannot be transmitted within the required timeframe. The time of transmission of the document is unknown.	A93
Document cannot be processed by receiving system	The receiving system cannot process that document in question.	A94
Complementary information	Additional text is provided in order to further explain a condition, for example to provide details of an outage.	A95
Technical constraint	A technical constraint has been applied.	A96
Force majeure curtailment	Curtailment due to Force Majeure. A code that enables the identification of the curtailment reason for settlement purposes.	A97
Network security curtailment	Curtailment due to network security reasons A code that enables the identification of the curtailment reason for settlement purposes.	A98
Auction cancelled	The auction has been cancelled.	A99
Incomplete document	The document is incomplete and cannot be processed.	B01
Accounting Point (tie-line) Time Series missing	The document is incomplete as a time series for an accounting point is missing.	B02
Meter data Time series missing	The document is incomplete as a time series for meter data is missing.	B03
Estimated values not allowed in first transmission	The document is in its initial form and estimated values are not allowed.	B04
No quantity values allowed for a quality that is not available	No quantity values are allowed for a quality that is not available.	B05
Time series accepted	Time series accepted.	B06
Auction without bids being entered	The auction has terminated without any bids being submitted. The ReasonText may provide the identification of the auction in question.	B07
Data not yet available	It is not possible to perform the necessary action since the required data for this action is not yet available.	B08
Bid not accepted	The bid in question has not been accepted.	B09
Initiator area problem	The problem concerns the initiator area.	B10
Cooperating area problem	The problem concerns the cooperating area.	B11
Communication status currently active	The status within the system indicates that the communication capability is currently active.	B12
Communication status currently inactive	The status within the system indicates that the communication capability is currently inactive.	B13
Communication status currently restricted	The status within the system indicates that the communication capability is currently restricted.	B14
Problem associated with both areas	The problem concerns both areas.	B15
Tender unavailable in MOL list	A tender that has been requested is no longer available in the MOL.	B16
Price based on preliminary exchange rate	The price is based on a preliminary exchange rate.	B17
Failure	A failure has occurred.	B18
Foreseen maintenance	Maintenance is foreseen.	B19
Shutdown	A shutdown has occurred.	B20
Official exchange rate approved	The official exchange rate has been approved.	B21
System regulation	The information provided regards a regulation for system purposes.	B22
Frequency regulation	The information provided regards a regulation for frequency purposes.	B23
Load flow overload	Situation in the grid, where loading of a certain grid element, e.g. overhead line, is above defined technical limits.	B24
	A TSO activity to maintain an acceptable voltage profile throughout the network. This is achieved by	

Voltage level adjustment	balancing of the respective reactive power requirements of the network and the customers.	B25
Emergency situation curtailment	Curtailment due to emergency situation. A code that enables the identification of the curtailment reason for settlement purposes.	B26
Calculation process failed	The calculation has not been performed.	B27
No capacity constraint impact on the market	The market position is such as no capacity constraint is applied to limit the cross border exchanges.	B28
Special Condition	Special condition need to be fulfilled.	B29
Unverified	Missing or not validated data.	B30
Verified	Data has successfully passed the verification process.	B31
CGM inconsistency	Describes an element which was not found in the CGM.	B32
Network dictionary inconsistency	Describes an element which was not found in the network dictionary.	B33
Capacity reduced by TSO	Describes a capacity that was reduced by a TSO.	B34
Overload	Describes an N-k or N state overload.	B35
GLSK limitation	Describes a situation in which there is not enough power adjustment in the GLSK file to cover the capacity.	B36
Voltage constraint	Describes an N-k or N state voltage violation.	B37
Angle constraint	Describes an N-k or N state angle violation.	B38
Stability	Describes a situation in which the dynamic behaviour of the network violated.	B39
Loadflow divergence	Describes a network situation in which the provided capacity values are part of a load flow divergence situation.	B40
Exclusion for SoS reasons	This is the adjustment applied to the capacity of a branch to have a minimum RAM (Remaining Available Margin) available for commercial exchanges.	B41
Constraint by the market	A constraint due to market restrictions.	B42
Ordinary	The contingency is ordinary (methodology for coordinating operational security analysis, article 6).	B43
Exceptional	The contingency is exceptional (methodology for coordinating operational security analysis, article 6).	B44
Out of range	The contingency is out of range (methodology for coordinating operational security analysis, article 6).	B45
Internal congestion	A temporary congestion within a bidding zone or scheduling area.	B46
Operational security constraints	Operational security constraints identified by TSOs.	B47
Estimated value	Describes a situation where a calculation process has failed and extrapolated or interpolated values will be applied.	B48
Balancing	Activated for balancing purposes.	B49
Values shared	Values of this time series are also valid for counter-party.	B50
Outside price limits	The offered price is not within the valid limits.	B51
Previous timeframe data	In case of processing issue, sent data are based on the previous timeframe.	B52
MOL merging succesful	The merging of the Merit Order List has been successfully processed.	B53
MOL merging failed	The merging of the Merit Order List has failed.	B54
Because of redispatching	Because of redispatching according to Article 2(26) of Commission Regulation (EU) 543/2013	B55
Because of countertrading	Because of countertrading according to Article 2(13) of Commission Regulation (EU) 543/2013	B56
Because of other remedial action	Not available because of any remedial action.	B57
Insufficiency of reserves	The reserve capacity is not sufficient.	B58
Unavailability of reserve providing unit	The unit providing the reserve is subject to technical unavailability.	B59
Unavailability of automatic protection systems	Unavailability of tools to detect predetermined system conditions that have a high probability of causing unusual stress on the power system, for which pre-planned remedial action is considered necessary or for which automatic protective actions may be triggered such as special protection schemes or automatic load shedding.	B60
Physical cable or converter restrictions	Limitation due to physical cable or converter restrictions.	B61
Constraints in controller systems	Limitation due to constraints in controller systems.	B62
Adjusted because of expected violation of operational security	The capacity is adjusted because of an expected violation of operational security limits of physical transmission assets.	B63
Adjusted because already considered remedial actions are assessed as not sufficient	The capacity is adjusted because the remedial actions were assessed as not sufficient or ineffective to avoid the expected violation of operational security limit(s).	B64
Time series partially accepted	The time series is partially accepted.	B65
Demand fully netted	Demand was fully netted against other demand(s).	B66
Bid activated in same direction	One or several bids were activated in the same direction within the same uncongested area as the demand.	B67
Optimization in progress	The activation optimisation framework is being executed.	B68
Wrong energy market timeframe	The given energy market timeframe is invalid or wrong.	B69
Message partially accepted	The detailed transactions of the received document are partially accepted. It is necessary to look at the detailed (transaction) level to determine if the transaction is accepted, rejected etc.	B70
(Explicit) Permission from the customer	Based on the regulation (EU) 2016/679 (GDPR) Article 6, Lawfulness of processing: The data subject has given consent to the processing of his or her personal data for one or more specific purposes.	B71
In execution of a contract	Based on the regulation (EU) 2016/679 (GDPR) Article 6, Lawfulness of processing: Processing is	

with the customer (Execution of contract)	necessary for the performance of a contract to which the data subject is party or in order to take steps at the request of the data subject prior to entering into a contract.	B72
Legal obligations	Based on the regulation (EU) 2016/679 (GDPR) Article 6, Lawfulness of processing: Processing is necessary for compliance with a legal obligation to which the controller is subject.	B73
In protection of vital interests of the customer	Based on the regulation (EU) 2016/679 (GDPR) Article 6, Lawfulness of processing: Processing is necessary in order to protect the vital interests of the data subject or of another natural person.	B74
In order to perform a duty of common (national) interest	Based on the regulation (EU) 2016/679 (GDPR) Article 6, Lawfulness of processing: Processing is necessary for the performance of a task carried out in the public interest or in the exercise of official authority vested in the controller.	B75
Justified interest of the dataprocessee	Based on the regulation (EU) 2016/679 (GDPR) Article 6, Lawfulness of processing: Processing is necessary for the purposes of the legitimate interests pursued by the controller or by a third party, except where such interests are overridden by the interests or fundamental rights and freedoms of the data subject which require protection of personal data, in particular where the data subject is a child	B76
Fulfilment of purpose	The purpose of the object or action has been fulfilled.	B77
Reach of end timestamp	The object or action has reached its end timestamp.	B78
Revocation	Revocation of the action or object.	B79
Termination	Termination of the action or object.	B80
No valid grid access contract	There is no valid grid access contract for a specific metering point. There is no valid grid access contract for a specific metering point.	B81
Customer move-out	A customer has moved out.	B82

BusinessTypeList

The exact business nature identifying the principal characteristic of a time series.

Literal	Description	Code
Production	The nature of the business being described is production details.	A01
Internal trade	The nature of the business being described is internal trade details.	A02
External trade explicit capacity	The nature of the business being described is external trade details between two areas with limited capacity requiring a capacity agreement identification.	A03
Consumption	The nature of the business being described is consumption details.	A04
External trade total	The nature of the business being described is external trade total.	A05
External trade without explicit capacity	The nature of the business being described is external trade details between two areas without requiring capacity allocation information.	A06
Net Production / Consumption	Net production/consumption - where signed values will be used. With the following rules: In area=Out area, In party=Out party, + means production and - means consumption.	A07
Net internal trade	Net internal trade - where the direction from out party (seller) to in party (buyer) is positive and the opposite direction is negative (with minus signs).	A08
IPP (Independent Power Producer)	A time series concerning the production schedule from an IPP.	A09
Tertiary control	A time series concerning tertiary reserve.	A10
Primary control	A time series concerning primary reserve.	A11
Secondary control	A time series concerning secondary reserve.	A12
Load profile	A time series concerning a load profile as calculated by a metered data aggregator.	A13
Aggregated energy data	A time series concerning adjusted metered readings received from a metered data collector and aggregated and validated by a metered data aggregator.	A14
Losses	A time series concerning losses that have been calculated for a tieline or an area.	A15
Transits (CBT)	A time series concerning inter area transit flows determined for CBT requirements.	A16
Settlement deviation	A time series concerning the imbalance energy calculated by an imbalance settlement responsible.	A17
Technical constraint deviation	A time series defining the imbalance between schedules accepted by the system operator due to technical constraints and schedules declared by the balance responsible party.	A18
Balance energy deviation	A time series defining the imbalance between the schedule of a balance responsible party that has been corrected by the system operator after using balance energy bids and the schedule that was accepted by the system operator due to technical constraints.	A19
Imbalance volume	A time series defining the imbalance between the actual meter readings and the schedule of the balance responsible party corrected by the system operator after using balance energy bids.	A20
Unintended energy	A timeseries concerning the volume of an unintended cross-border exchange of energy.	A21
Frequency control	A time series concerning primary and secondary reserve.	A22
Balance management	A time series concerning energy balancing services.	A23
Total trade	A time series concerning the total of both the internal and external trades.	A24
General Capacity Information	A time series providing the total capacity available on a TSO border.	A25
Available transfer capacity (ATC)	Available transfer capacity for cross-border exchanges.	A26
Net transfer capacity (NTC)	Net transfer capacity for cross-border exchanges.	A27
Control Area Program	A time series providing the total exchanges between two TSOs (including the commercial transactions, the compensation program and the losses compensation program). Note this definition might change when UCTE brings forward its coding requirements.	A28
Already allocated capacity (AAC)	The already allocated capacity is the total amount of allocated transmission rights.	A29

Internal inter area trade	A trade that occurs between internal areas within a market balance area.	A30
Offered Capacity	The time series provides the offered capacity.	A31
Capacity transfer notification	The time series provides information concerning the notification of the transfer of capacity to another market participant.	A32
Authorised AAC	The time series in question provides the amount of transmission capacity rights to be nominated.	A33
Capacity rights	The time series in question provides the capacity rights allocated for a given border.	A34
Minimum authorised AAC	The time series in question provides the minimum amount of transmission capacity rights to be nominated.	A35
Maximum authorised AAC	The time series in question provides the maximum amount of transmission capacity rights to be nominated.	A36
Installed generation	The time series in question provides the installed generation.	A37
Available generation	The time series in question provides the available generation.	A38
Interconnection Trade Responsible Designation	The Time series in question provides the designation of the ITR that may nominate the capacity in question.	A40
Released AAC	The already allocated capacity (AAC) that has been released for resale.	A41
Requested capacity (with price)	The time series in question provides information concerning the requested capacity including price information.	A42
Requested capacity (without price)	The time series in question provides information concerning the requested capacity but excludes price information.	A43
Compensation program	Compensation of unintentional deviation is performed by exporting to / importing from the interconnected system during the compensation period by means of schedules as calculated during the accounting of unintentional deviations.	A44
Schedule activated reserves	The cross border or internal reserves that are to be activated through schedule nomination.	A45
System Operator redispatching	The cross border redispatching between System Operators that are to be activated through schedule nomination.	A46
Market capacity price	The price of the capacity offered on a given market.	A47
Market capacity price differential	The difference between the price of capacity in a Market Balance Area receiving the capacity (In Area) and the price of capacity in a Market Balance Area providing the capacity (Out Area), i.e. In Area Price - Out Area price.	A48
Inflow	The volume of water that flows into a reservoir in a given interval.	A49
Water extraction	The volume of water that can be extracted from a reservoir in a given interval.	A50
Turbined water	The volume of water that can be turbined in a plant in a given interval.	A51
Water spillage	The volume of water that is not turbined going through the spillway in a given interval.	A52
Planned maintenance	Maintenance has been planned for the object in question with a forecast ending date.	A53
Unplanned outage	An unplanned outage has occurred on the object in question.	A54
Use it Or Sell it (UIOSI) pricing	The time series provides information on the capacity resold in the "use it or sell it" process and its corresponding price.	A55
Compensation for auction cancellation where capacity is for resale	The time series provides information on the compensation of the capacity for resale following an auction cancellation.	A56
Resale pricing	For each Physical Transmission Rights holder, this document contains the resold capacity and its corresponding price.	A57
Curtailed capacity compensation	The time series provides information to compensate a party when curtailment is applied on the capacity obtained in a previous auction, resale or transfer.	A58
Use it Or Sell it (UIOSI) compensation	The time series provides information on the compensation for the capacity following an auction cancellation.	A59
Minimum possible	The time series provides a schedule of minimum possible values for a Resource Object. The nature of the flow could be defined by the attribute Direction.	A60
Maximum available	The time series provides a schedule of maximum available values for a Resource Object. The nature of the flow could be defined by the attribute Direction.	A61
Spot price	The time series provides the market spot prices from an auction.	A62
Minimum ATC	The Available Transmission Capacity that must be guaranteed because of regulatory constraints.	A63
Meter Measurement data	The data as provided for a meter measurement source.	A64
Accounting Point Relevant data	The metered data that is to be considered relevant for accounting purposes.	A65
Energy flow	Energy flow information.	A66
Power plant energy Schedule	Energy flow scheduled for a power plant.	A67
Compensation Requirements for the compensation period	The time series provides the compensation requirements for a given compensation period.	A68
Market coupling results	The time series provides the results of a market coupling auction.	A69
Production, unavailable	Production capacity that normally would be available, but due to maintenance or similar is temporarily unavailable.	A70
Supplementary available generation	The supplementary generation that is available.	A71
Interruptible consumption	The consumption that may be interrupted on request.	A72
Summarised Market Balance Area Schedule	A time series providing the total exchanges based on commercial transactions between two Market Balance Areas.	A73
Load Frequency		

Control Program Schedule	A time series providing the schedule information for the Load Frequency Control Program.	A74
Timeframe Independent Schedule	A time series providing the total exchanges of Timeframe Independent Schedules between two System Operators.	A75
Consumption curtailment	A time series providing the amount of voluntary consumption curtailed by the energy supplier of an end-consumer.	A76
Production, dispatchable	The nature of the business being described is dispatchable production details, i.e. generation output that can be changed by a request (activation order) of the TSO according with the applicable Market Rules.	A77
Consumption, dispatchable	The nature of the business being described is dispatchable consumption details, i.e. consumption output that can be changed by a request (activation order) of the TSO according with the applicable Market Rules.	A78
Production, non-dispatchable	The nature of the business being described is non-dispatchable production details, i.e. generation output that cannot be modified by an activation order.	A79
Consumption, non-dispatchable	The nature of the business being described is non-dispatchable consumption details, i.e. consumption output that cannot be modified by an activation order.	A80
Total Transfer Capacity (TTC)	The Total Transfer Capacity is the maximum exchange program between two areas compatible with operational security standards applicable at each system if future network conditions, generation and load patterns were perfectly known in advance.	A81
Mutual Emergency Assistance Service (MEAS)	The cross border Mutual Emergency Assistance Service between System Operators that are to be activated through schedule nomination.	A82
Auction cancelation	The time series covers auction cancellation right.	A83
Nomination curtailment	The time series covers nomination curtailment rights	A84
Internal redispatch	Redispatch to relieve Market Balance Area internal congestion.	A85
Control area balance energy	A sum of secondary, tertiary control as well as other energy that was used to balance a control area.	A86
Balancing energy price	Price of energy usedto balance.	A87
Economised secondary reserve	The activated secondary reserve that had been economised due to pooled reserve management.	A88
Spinning reserve	The extra generating capacity that is available by increasing the production of generators that are already connected to the power system.	A89
Solar	The business being described concerns solar power.	A90
positive forecast margin	The business being described concerns a positive forecast margin.	A91
Negative forecast margin	The business being described concerns a negative forecast margin.	A92
Wind generation	The business being described concerns wind generation.	A93
Solar generation	The business being described concerns solar generation.	A94
Frequency containment reserve	The business being described concerns frequency containment reserve.	A95
Automatic frequency restoration reserve	The business being described concerns automatic frequency restoration reserve.	A96
Manual frequency restoration reserve	The business being described concerns manual frequency restoration reserve.	A97
Replacement reserve	The business being described concerns replacement reserve.	A98
Financial information	The business being described concerns financial information.	A99
Interconnector network evolution	The business being described concerns interconnector network evolution.	B01
Interconnector network dismantling	The business being described concerns interconnector network dismantling.	B02
Counter trade	The business being described concerns counter trades.	B03
Congestion costs	The business being described concerns congestion costs.	B04
Capacity allocated (including price)	The business being described concerns capacity allocation and includes price information.	B05
DC link constraint	The business being described concerns DC link constraints.	B06
Auction revenue	The business being described concerns auction revenue.	B07
Total nominated capacity	The business being described concerns the total nominated capacity.	B08
Net position	The business being described concerns net position.	B09
Congestion income	The business being described concerns congestion income.	B10
Production unit	The business being described concerns a production unit.	B11
Rounded market coupling results	Rounded outputs of the market coupling to be sent to TSOs and Market Participants.	B12
AllocationRevenue	The time series provides information on the revenue generated by the allocations.	B13
Production deviation	A time series concerning the imbalance energy between the metered production and the schedules calculated by an imbalance settlement responsible.	B14
Consumption deviation	A time series concerning the imbalance energy between metered consumption and the forecasted consumption calculated by an imbalance settlement responsible.	B15
Transmission asset	The business being described concerns atransmission asset.	B16
Consumption unit	The business being described concerns a consumption unit.	B17
In-feed ATC	Available Transfer Capacity at the in-feed side of a DC tieline.	B18
Out-feed ATC	Available Transfer Capacity at the out-feed side of a DC tieline.	B19
Balance up regulation price	A time series concerning balance regulation market prices for up regulation.	B20
Balance down regulation price	A time series concerning balance regulation market prices for down regulation.	B21

Main direction	A time series concerning the direction of balance regulations.	B22
Consumption imbalance price	A time series concerning imbalance prices for consumption.	B23
Production sales imbalance price	A time series concerning imbalance prices for production sales.	B24
Production purchase imbalance price	A time series concerning imbalance prices for production purchase.	B25
Average balance price between MBAs	A time series concerning the average prices between Market Balance Areas.	B26
Pumped	A time series concerning the electricity consumption related to pumping.	B27
Large installation consumption	A time series concerning consumption from large installation.	B28
Metering Grid Area (MGA) imbalance	A time series concerning imbalance between reported consumption, production and exchange in a Metering Grid Area.	B29
HVDC Link settings	The time series in question provides HVDC Link settings.	B30
Transmission Reliability Margin (TRM)	A time series concerning Transmission Reliability Margin (TRM).	B31
Imbalance component for a pool	This information is used to provide to a pool manager the combined imbalance of all the pool participants.	B32
Area Control error (ACE)	The sum of the instantaneous difference between the actual and the set-point value of the measured total power value and Control Program including Virtual Tie-Lines for the power interchange of a LFC Area or a LFC Block and the frequency bias given by the product of the K-Factor of the LFC Area or the LFC Block and the Frequency Deviation.	B33
Area Control Error after Imbalance Netting	A time series concerning the Area Control Error after applying the imbalance netting energy correction.	B34
Implicit and explicit trade total	The sum of cross border schedules based on implicit and explicit trades including long term, yearly, monthly, weekly, daily processes.	B35
Production units own consumption	The consumption of one or more production units.	B36
Constraint situation	The timeseries describes the constraint situation for a given TimeInterval. A constraint situation can be: - composed of a list of network elements in outage associated for each outage to a list of network elements on which remedial actions have been carried out accordingly to contingency process - or it can be an external constraint.	B37
Initial domain	The timeseries describe the full flow based domain for a given TimeInterval. Critical network elements are displayed in details and their impact on the market is quantified.	B38
Flow based domain adjusted to long term schedules	The timeseries describe the full flow based domain for a given TimeInterval adjusted to the latest update of the schedules. Critical network elements are displayed in details and their impact on the market is quantified.	B39
Network element constraint	The timeSeries describes limiting elements which are overloaded.	B40
Calculation opposition (Red Flag)	The timeSeries describes a party who is opposed to the calculation result and imposes its transfer capacity value.	B41
Base case proportional shift key	The GSK or LSK are proportional to the base case generation or load.	B42
Proportional to participation factors shift key	The GSK or LSK are proportional to the participation factors.	B43
Proportional to the remaining capacity shift key	The GSK is proportional to the remaining available capacity.	B44
Merit order shift key	The GSK is proportional to a merit order list.	B45
Wind speed	The TimeSeries provides information on the wind speed.	B46
Wind direction	The TimeSeries provides information on the wind direction.	B47
Solar irradiance	The TimeSeries provides information on the power per unit area produced by the sun in the form of electromagnetic radiation.	B48
Air temperature	The TimeSeries provides information on the air temperature.	B49
Cloudiness	The TimeSeries provides information on the cloudiness, i.e. the level of coverage of the sky with clouds.	B50
Air humidity	The TimeSeries provides information on the level of humidity of the air.	B51
Atmospheric pressure	The TimeSeries provides information on the atmospheric pressure.	B52
Precipitation	The TimeSeries provides information on the amount of rain, snow, etc. that falls on the ground.	B53
Network constraint situation that constraints the market	The TimeSeries describes the network elements, that constraints the market, to be taken into account to simulate a network constraint during the network load flow studies. The network situation includes the contingencies, the remedial actions, the monitored network elements and the potential additional constraints.	B54
Contingency	The TimeSeries describes the network elements part of the contingency to be simulated for a given TimeInterval.	B55
Remedial Action	The TimeSeries describes a set of remedial actions for a given TimeInterval.	B56
Monitored Network Element	The TimeSeries describes the network elements to be monitored during the network load flow studies.	B57
Busbar	The TimeSeries describes the network elements that composed a busbar.	B58
Network Element	The TimeSeries describes network elements.	B59
SPS	The TimeSeries describes the network elements managed by a Special Protection System (automation).	B60
Aggregated netted external market schedule	The aggregated netted external market schedules for a given border.	B61

Aggregated netted external TSO schedule	The aggregated netted external TSO schedules for a given border.	B62
Aggregated netted external schedule	The aggregated netted external schedules for a given border.	B63
Netted area AC position	The AC position for a given area.	B64
Netted area position	The netted aggregation of all AC external schedules of an area plus the aggregated External Netted Schedules of related HVDC links of an area.	B65
Interconnection shift key	The shift key series describes the amount of power to be shifted from a border area.	B66
DC flow with losses	DC flow with losses refers to the values at the importing end of the DC line.	B67
DC flow without losses	DC flow without losses refers to the values at the exporting end of the DC line.	B68
minimum value of netted area position	That value which a netted area position must not fall below for a given area.	B69
maximum value of netted area position	That value which a netted area position must not exceed for a given optimisation area.	B70
maximum value of DC flow	That value which a balanced DC flow must not exceed for a given DC line on exporting end. When aligning DC flows CGMA algorithm will respect this constraint.	B71
minimum value of DC flow	That value which a balanced DC flow must not fall below for a given DC line on exporting end. Currently this business type is only included for consistency reasons. It is always set to 0. This constraint might, however, be used in future. When aligning DC flows the CGMA algorithm will respect this constraint.	B72
indicative AC flow	It is the hypothetical flow on the aggregate of all AC tie lines of an electrical border between two optimisation areas. It results from the adjustments to the preliminary netted area positions of all optimisation areas made by the CGMA algorithm. Indicative AC flows are an artefact of the CGMA algorithm, and do not correspond to physical flows	B73
Offer	The time series provides an offer to provide reserves.	B74
Need	The time series provides a requirement for reserves.	B75
Opportunity costs or benefits	The time series describes any opportunity costs or benefits.	B76
Financial compensation or penalties	The time series describes any financial compensation or penalties	B77
Global radiation	The total short-wave radiation from the Global radiation is the total short-wave radiation from the sky falling onto a horizontal surface on the ground. It includes both the direct solar radiation and the diffuse radiation resulting from reflected or scattered sunlight.	B78
Diffuse radiation	Radiation resulting from reflected or scattered sunlight.	B79
Direct solar radiation	Radiation resulting from direct sunlight	B80
Outage (OUT)	Outage process: Element is out of operation due to planned maintenance or due to an unplanned/forced outage. Outage may be used as a synonym for unavailability.	B81
Special switching state (SSS)	Outage Process: This state applies to grid elements which are in operation in an exceptional state (e.g. separated nodes operation).	B82
Testing (TEST)	Outage process: TESTING means any element status is possible - ON or OUT. This status applies either between first connection and final commissioning of the relevant asset, or directly following maintenance of the relevant asset.	B83
Auxiliary busbar operation	Outage process: Element is in operation but connected via auxiliary busbar	B84
Automatic reclosing	Outage process: Protection function Automatic reclosing is switched off for electric line	B85
Busbar protection	Protection function busbar protection is switched off	B86
Phase Shift Angle	The maximum phase shift angle allowed between two network elements.	B87
Base Case Network Situation	The TimeSeries describes the network elements to be taken into account to simulate a base case network situation during the network load flow studies, without any contingency.	B88
Inter-TSO assistance	Cross border assistance schedule between TSOs not interconnected directly.	B89
FlexibleNeed	The business type indicates that the need is optional.	B90
GLSK Limitation	A constraint related to a GLSK maximum or minimum limitation in the production or/and consumption shift.	B91
Capacity ramping limitation	A constraint related to a ramping limitation on the capacity offered at a given border.	B92
interconnector capacity	The maximum capacity that can be exchanged on an interconnector, excluding external factor on both ends.	B93
Must Run	A time series concerning must run generation.	B94
Procured capacity	An accepted offer of balancing capacity.	B95
Used capacity	The used cross-zonal balancing capacity.	B96
Estimated costs	Estimated costs of the process.	B97
Estimated benefits	Estimated benefits of the process.	B98
Load Shedding	A time series concerning a load shedding used to avoid failure of the power system.	B99
Remaining Capacity	A time series concerning the remaining capacity.	C01
Indicator of generation capacity adequacy	Indicator of adequacy, it indicates if there is final generation remaining capacity after SMTA calculation.	C02
Income from price divergence without congestions	The time series describes income due to price divergence without congestion between bidding zones.	C03
Push-button	The cross-border Push-button service between System Operators.	C04
Intertripping	The cross-border Intertripping service between System Operators.	C05
Emergency instruction	The cross-border Emergency instruction service between System Operators.	C06
Ramp management	The schedule resulting from cross-border Ramp management service between System Operators.	C07

Profile smoothing	The schedule resulting from cross-border Profile smoothing service between System Operators.	C08
Emergency reallocation deselection	The schedule resulting from cross-border Emergency reallocation deselection service between System Operators.	C09
SO-SO-trade	The generic cross border trade between System Operators.	C10
Production reduction	A time series providing the volume of production reduced by an energy provider / producer / supplier.	C11
Maximum power exchange	The timeseries provides the maximum admissible power flow between two bidding zones respecting operational security limits taking into account N-1 criterion.	C12
Maximum power exchange after remedial actions	The timeseries provides the maximum admissible power flow between two bidding zones after remedial actions.	C13
Network constraint situation that cannot limit the market	The TimeSeries describes the network elements, that cannot limit the market, to be taken into account to simulate a network constraint during the network load flow studies. The network situation includes the contingencies, the remedial actions, the monitored network elements and the potential additional constraints.	C14
Flat participation for all generators or loads	Flat GSK factors of all generators or loads, independently of the size.	C15
Proportional to installed capacity of generators	Generators participate relative to their maximum (installed) capacity (MW).	C16
Market price and total volume	A time series concerning market price and total volume.	C17
Import price	A time series concerning import price (the volume-weighted price average of all accepted bids).	C18
Capacity allocated (excluding price)	The business being described concerns capacity allocation and excludes price information.	C19
Common Grid Model Equipment	The timeseries provides equipment related to the Common Grid Model (CGM).	C20
Exchanged balancing reserve capacity	The balancing reserve capacity exchanged between areas.	C21
Shared balancing reserve capacity	The balancing reserve capacity shared between areas.	C22
Share of reserve capacity	A time series concerning the share of reserve capacity.	C23
Actual reserve capacity	A timeseries concerning actual reserve capacity.	C24
K-factor	K-factor as stated in the SO GL Art. 2 (45). It is also known as Frequency Bias.	C25
Frequency Containment Reserve-Normal (FCR-N)	FCR-N is a reserve that is automatically activated in both directions around a set point when the frequency varies between 50.10 Hz and 49.90 Hz after an imbalance.	C26
Frequency Containment Reserve-Disturbance (FCR-D)	FCR-D is a reserve that is automatically activated when the frequency falls below 49.90 Hz or rises above 50.1 Hz after an imbalance.	C27
Internal trade difference	A time series concerning internal trade difference, within an area, such as a Bidding Zone or Scheduling Area. The internal trade difference is the difference between trades reported from an out party (seller) and an in party (buyer).	C28
Small scale production	Production from small scale production plants.	C29
System price	The system price is an unconstrained market clearing reference price. It is calculated without any congestion restrictions by setting capacities to infinity.	C30
Wind gust	An increase in the speed of the wind lasting for a short period.	C31
Area imbalance	A time series concerning imbalance between planned consumption, production and exchange in an Area.	C32
Unintended energy price	A timeseries concerning the price of the unintended cross-border exchange of energy.	C33
Frequency containment process energy	A timeseries containing the volume of energy resulting from the frequency containment process.	C34
Frequency containment process energy price	A timeseries containing the energy price from the frequency containment process.	C35
Ramping period energy	A timeseries containing the volume of energy exchanged as a result of ramping between different ANES values.	C36
Ramping period energy price	A timeseries containing the price of the energy exchanged as a result of ramping between different ANES values.	C37
Frequency deviation	A timeseries concerning the difference between the actual and the nominal frequency of a synchronous area.	C38
Day-Ahead market price	A timeseries concerning Day-Ahead market prices.	C39
Conditional bid	Standard product bid that is conditional on bids submitted outside of common platform.	C40
Thermal limit	The current causing a given network element to work outside of the range of safe operating temperatures.	C41
Frequency Limit	A constraint related to the containment of frequency deviations within a given area.	C42
Voltage limit	The maximum or minimum permissible voltage within normal operation state of a given network element.	C43
Current limit	The maximum permissible current within normal operation state of a given network element.	C44
Short circuit current limit	The maximum permissible short-circuit current within normal operation state of a given network element.	C45
Dynamic stability limit	A maximum permissible load ensuring the control of oscillations in the grid and avoiding the loss of synchronism.	C46
Disconnection	A timeseries describing disconnection of a TSO from a common platform.	C47
Intended energy with positive price	A timeseries concerning the amount of intended energy with prices higher than zero (and including zero).	C48
Intended energy with negative price	A timeseries concerning the amount of intended energy with prices lower than zero (excluding zero).	C49
Decoupling	A time series describing decoupling of an area.	C50
Resource capacity	A timeseries containing information about resource capacity units.	C51

unit		
Resource entry capacity data	A timeseries containing the resource capacity that can be allocated to an eligible resource capacity operator from another area.	C52
Resource capacity obligation data	A timeseries containing the resource capacity operator obligation to guarantee delivery.	C53
Available energy	A timeseries concerning the available energy.	C54
Production curtailment	A timeseries concerning the curtailment of production.	C55
Rounding error	A timeseries describing a rounding error.	C56
Metered frequency	The timeseries provides information about metered frequency.	C57
Adjusted TTC to the nominal criteria	The exchange program between two areas which guarantees that the Margin Available for Cross-Zonal Trade (MACZT) fulfils the nominal criteria at least on the most limiting Critical Network Element with Contingency (CNEC) which limits the transfer capacity.	C58
Adjusted TTC to the nominal criteria with TSOs limitation	The exchange program between two areas which allows the Margin Available for Cross-Zonal Trade (MACZT) on, at least, the most limiting Critical Network Element with Contingency (CNEC) to get closer to the nominal criteria fulfilment with a limited impact on the rest of the network.	C59
Frequency deviation larger than standard deviation	Total time in which the absolute value of the instantaneous frequency deviation was larger than the standard frequency deviation.	C60
Frequency deviation larger than maximum deviation	Total time in which the absolute value of the instantaneous frequency deviation was larger than the maximum instantaneous frequency deviation.	C61
Frequency deviation not returned to 50%	Number of events in which the absolute value of the instantaneous frequency deviation of the synchronous area exceeded 200 % of the standard frequency deviation as stated in SO GL (EU) regulation Art 131.1.a.vi.	C62
Frequency deviation not returned to restoration range	Number of events in which the absolute value of the instantaneous frequency deviation of the synchronous area exceeded 200 % of the standard frequency deviation.	C63
Frequency deviation outside recovery range	Number of events for which the absolute value of the instantaneous frequency deviation was outside of the frequency recovery range.	C64
Frequency	A time series describing measurement frequency.	C65
Mean value	A time series describing mean values.	C66
Standard deviation	A time series describing standard deviation.	C67
Percentile	A time series describing percentiles.	C68
Measured frequency resolution	A time series describing the resolution of a measured frequency.	C69
Accuracy	A time series describing measurement accuracy.	C70
FRCE outside level 1 range	The number of time intervals in which the average value of the FRCE was outside the Level 1 FRCE range as stated in SO GL. (EU) regulation Art 131.1.b.i.	C71
FRCE outside level 2 range	The number of time intervals in which the average value of the FRCE was outside the Level 2 FRCE range as stated in SO GL (EU) regulation. Art 131.1.b.i.	C72
FRCE exceeded 60% of FRR capacity	The number of events for which the FRCE exceeded 60 % of the reserve capacity on FRR as stated in SO GL (EU) regulation Art 131.1.b.ii.	C73
FRCE exceeded steady state deviation	The number of events for which the absolute value of the FRCE exceeded the maximum steady-state frequency deviation.	C74
FRCE calculation and accuracy descriptor	A time series describing how FRCE is calculated and its accuracy.	C75
Forecasted capacity	A time series describing forecasted capacity.	C76
Minimum available capacity	A time series describing minimum available capacity.	C77
Average available capacity	A time series describing average available capacity.	C78
Maximum available capacity	A time series describing maximum available capacity.	C79
Frequency and accuracy descriptor	A time series describing how system frequency and accuracy are determined.	C80
Long-Term internal redispatch	A time series describing long-term redispatch to relieve Scheduling Area internal congestion.	C81
Other unavailability	This is an unplanned unavailability. Not considered by market participants as a planned maintenance.	C82
Faster than standard FAT	Bids that can support a ?Full Activation Time? (FAT) that is faster than standard FAT. Fast activation can be done for bids with activation time shorter than the minimum requirement for the standard product. When circumstances call for it, the TSO can order activation of such bids on a shorter notice.	C83
Faster than standard deactivation time	Fast deactivation can be done for bids with activation time shorter than the minimum requirement for the standard product. When circumstances call for it, the TSO can order activation of such bids on a shorter notice.	C84
Slower than standard FAT	Bids that can support a Full Activation Time (FAT) that is slower than standard FAT.	C85
Remedial action cost summary	A timeseries summarizing all incurred costs and/or revenues per party related to the activated remedial actions eligible to Cost Sharing.	C86
Settlement result	A timeseries representing the cost sharing settlement results between parties (i.e. which party will be paying and/or receiving money for the concerned period).	C87
Reserved cross zonal capacity	The reserved cross zonal capacity for system operator needs (EBGL art. 38(1)(b) and art. 41.).	C88
Energy reserves	A timeseries describing energy reserves.	C89
Combined dynamic constraint	The time series describes a combined dynamic constraint, being defined as a limit on the sum of power flows on a set of network elements or partial flows on a set of network elements for the purpose to respect dynamic stability limits.	C90

Anonymised constraint	The time series describes an anonymised constraint.	C91
Synchronous condenser activation	Activates a number of generators or other units in synchronous condenser mode.	C92
Permission administration details	A timeseries describing permission details such as authorisations, licenses or related regulatory requirements.	C93
Maximum capacity value of AC border flow	Maximum flow on an AC border between two areasÂ (capacity)	C94
Initial forecast value of AC border flow	Initial flow on an AC border between two areas (forecast)	C95

EnergyProductTypeList

The identification of the nature of an energy product such as power, energy, reactive power, etc.

Literal	Description	Code
Active power	The product of voltage and the in-phase component of alternating current measured in units of watts and standard multiples thereof.	8716867000016
Reactive power	The product of voltage and current and the sine of the phase angle between them, measured in units of voltamperes reactive and standard multiples thereof. (not used for planned schedules).	8716867000023
Active energy	The electrical energy produced, flowing or supplied by an electrical circuit during a time interval, being the integral with respect to time of instantaneous active power, measured in units of watt-hours, or standard multiples thereof.	8716867000030
Reactive energy	The integral with respect to time of reactive power (not used for planned schedules).	8716867000047
Capacitive Reactive energy	Capacitive reactive energy.	8716867000139
Inductive Reactive energy	Inductive reactive energy.	8716867000146
Water	For hydro power stations, this enables the identification of the quantity of water stored behind a dam (volume, head level, etc.), or the constraints in the flow of water.	8716867009911
Capacitive reactive power	Capacitive reactive power.	8716867000115
Inductive reactive power	Inductive reactive power.	8716867000122

MessageTypeList

The coded type of a document. The message type describes the principal characteristic of a document.
This enumeration is used in the XML instances based on IEC 62325.

Literal	Description	Code
Balance responsible schedule	A schedule which has been prepared by a balance responsible party providing planned schedule information.	A01
Allocated capacity schedule	A schedule which has been prepared by a capacity allocator providing allocated capacity.	A02
Balance area schedule	A schedule that provides the planned schedule information for a balance area.	A03
System Operator area schedule	A compilation of all external schedules concerning two System Operator areas or a connector concerning two System Operator of all balance responsible parties.	A04
Control block area schedule	A compilation of all the exchange programs of all control areas for one control block with all neighbouring control areas of a neighbouring control block.	A05
Coordination center area schedule	A compilation of the exchange programs of all exchange blocks divided into UCTE south and north.	A06
Intermediate confirmation report	An intermediate confirmation report that may be produced between final cutoffs.	A07
Final confirmation report	A final confirmation report that is produced after a final cutoff.	A08
Finalised schedule	A compilation of a set of schedules that have been finalized after a given cutoff.	A09
Regulation data report	A compilation of the time series employed on a given day to ensure the balance of the system.	A10
Aggregated energy data report	A compilation of the time series of all the meter readings or their equivalent for a given period.	A11
Imbalance report	The report containing the complete situation of a given period for a party and including the schedules, regulation data and actual or calculated readings.	A12
Interconnection Capacity	Document for cross-border capacity exchanges.	A13
Resource Provider Resource Schedule	A document providing the schedules for resource objects submitted by a resource provider.	A14
Acquiring System Operator Reserve Schedule	A document providing reserve purchases submitted by an Acquiring System Operator.	A15
Anomaly Report	A document providing anomaly information for the receiving party to correct.	A16
Acknowledgement Document	A document providing acknowledgement information.	A17
Confirmation report	A document providing confirmation information.	A18

Capacity for Resale	A document providing information about capacity for resale.	A19
Approved Capacity Transfer	A document to approve a capacity transfer.	A20
Capacity transfer notification	A document notifying a capacity transfer.	A21
Transmission rights portfolio	A document providing the portfolio of the transmission capacity rights of a market participant.	A22
Allocations	A document providing the capacity allocations for a border.	A23
Bid document	A document providing bid information.	A24
Allocation result document	A document providing the allocation results of an auction.	A25
Capacity document	A document providing capacity information.	A26
Rights document	A document providing transmission capacity rights information.	A27
Generation availability schedule	This document contains information related to energy availability.	A28
Cross border schedule	This document contains the cross border schedules for all the borders of a given country where energy is exchanged.	A30
Agreed capacity	The capacity agreed between parties.	A31
Proposed capacity	The capacity proposed for agreement between parties.	A32
System vertical load	The sum of all flows out of the transmission grid via directly connected transformers and lines to distribution grids and end consumers as known by the System Operator.	A33
Escalation document	A document which requesting the escalation of a situation.	A34
Trouble shooting document	A document providing trouble shooting information for the resolution of a problem.	A35
Deactivation document	A document providing deactivation information.	A36
Reserve tender document	The document that is used for the tendering for reserves within the ERRP process.	A37
Reserve Allocation Result Document	The document used to provide the results of a Reserve auction.	A38
SATCR activation	The document is used to provide the activation of reserves through the SATCR process.	A39
DATCR activation	The document is used to provide the activation of reserves through the DATCR process.	A40
Activation response	The document is used to provide a response to a request to activate reserves.	A41
Tender reduction	The document is used to provide information concerning the reduction of a previously submitted tender.	A42
MOL Document	The document is used to provide Merit Order List information.	A43
Price Document	The document is used to provide market price information.	A44
Measurement Value Document	The document is used to provide measurement information from measurement devices.	A45
SOAM Document	The document is used to provide system operator accounting data for matching.	A46
SOVA Document	The document is used to provide system operator validated accounting data.	A47
CCVA Document	The document is used to provide coordination center validated accounting data.	A48
Daily settlement document	The document is used to provide daily settlement information.	A49
Weekly settlement document	The document is used to provide weekly settlement information.	A50
Capacity Auction Specification Document	The document is used to provide capacity auction specification information.	A51
Market Coupling Results Document	The document is used to provide the results of a market coupling auction.	A52
Outage publication Document	The document is used to provide the outage information for publication.	A53
Forced generation outage Document	A document providing information on forced generation outages.	A54
Summarised Market Schedule	A compilation of all external schedules concerning two Market Balance Areas of all balance responsible parties.	A55
Compensation Program Schedule	A schedule that provides the schedule information for the compensation of unintended deviation.	A56
Load Frequency Control Program Schedule	A schedule that provides the schedule information for the Load Frequency Control Program of a Control Area or a Control Block.	A57
Timeframe Independent Schedule	A compilation of all external Timeframe Independent Schedules concerning two System Operators.	A58
Information request	An information request being made concerning some specific information.	A59
status request for a position independently from a specific process	A status request concerning the position of an object independently of any ongoing processes.	A60
Estimated Net Transfer Capacity	The estimated net transfer capacity for a given border.	A61
Compensation rights	The capacity rights that have been allocated as compensation.	A62
Redispatch notice	A notice to confirm the actions agreed between System Operators to resolve a congestion situation through redispatch.	A63
Tender reduction response	A response to a tender reduction request that provides corrections to the initial document.	A64
System total load	Total load', including losses without power used for energy storage, is equal to generation deducted with exports, added with imports and deducted with power used for energy storage.	A65
Final MOL	A document providing the information concerning the situation of the MOL at the end of an activation period.	A66
Resource Provider Schedule for production/consumption	A document providing the schedules for production/consumption for resource objects submitted by a resource provider.	A67
Installed generation	A document providing the installed generation per generation type.	A68

per type		
Wind and solar forecast	A document providing the forecast of wind and solar generation.	A69
Load forecast margin	A document providing the load forecast margin for a period.	A70
Generation forecast	A document providing the generation forecast for a period.	A71
Reservoir filling information	A document providing information concerning the filling of reservoirs.	A72
Actual generation	A document providing the actual generation for a period.	A73
Wind and solar generation	A document providing the generation of wind and solar energy for a period.	A74
Actual generation per type	A document providing the actual generation per generation type for a period.	A75
Load unavailability	A document providing the unavailability of units providing load on the network.	A76
Production unavailability	A document providing the unavailability of production units providing energy to the network.	A77
Transmission unavailability	A document providing the unavailability of transmission capacity on the network.	A78
Offshore grid infrastructure unavailability	A document providing the unavailability of an offshore grid infrastructure to the network.	A79
Generation unavailability	A document providing the unavailability of generation units providing energy to the network.	A80
Contracted reserves	A document providing the reserves contracted for a period.	A81
Accepted offers	A document providing the offers of reserves that have been accepted for a period.	A82
Activated balancing quantities	A document providing the quantities of reserves that have been activated for balancing.	A83
Activated balancing prices	A document providing the prices of the reserves that have been activated for balancing.	A84
Imbalance prices	A document providing the prices of reserves due to imbalance for a period.	A85
Imbalance volume	A document providing the volume of the imbalance for a period.	A86
Financial situation	A document providing the financial situation for reserves.	A87
Cross border balancing	A document providing the cross border balancing requirements for a period.	A88
Contracted reserve prices	A document providing the price of reserves contracted for a period.	A89
Interconnection network expansion	A document providing information on the expansion of the interconnection network.	A90
Counter trade notice	A document providing information on counter trades for a period.	A91
Congestion costs	A document providing the cost of congestion for a period.	A92
DC link capacity	A document providing the DC links for a period.	A93
Non EU allocations	A document providing allocations made to non EU member states.	A94
Configuration document	A document providing configuration information.	A95
Redispatch activation document	A document enabling the activation of a redispatch notice.	A96
Detailed activation history document	A document enabling a detailed history of activations.	A97
Aggregated activation history document	A document enabling an aggregated history of activations.	A98
HVDC Link constraints	A document providing the information concerning the maximum and minimum active power flow through each link can limited.	A99
HVDC Configuration	A document providing the information concerning the power set point. The mode in which the HVDC is managed.	B01
HVDC Schedule	A document providing the information for operating DC links.	B02
EIC code request	A document providing the information requesting a new EIC code.	B03
EIC code information	A document providing EIC information in a central registry exchange or information to an EIC participant.	B04
EIC code publication	A document providing EIC publication information in a web site publication of a limited set of information.	B05
Critical network element determination	A document providing all the elements necessary for the capacity coordinator to determine the transfer capacity and the associated critical network elements.	B06
Critical network element publication	A document providing all the elements necessary for the market information aggregator and TSOs to know the critical network elements which limit the transfer capacity.	B07
Flow based domain	A document providing the capacity domain and its limits available for the TSO.	B08
Flow based domain publication	A document providing the capacity domain and its limits available for the market.	B09
Flow based domain market impact publication	A document providing the capacity domain and its impacts on the market to be published.	B10
Anonymized flow based parameters publication	A document providing all the relevant flow based parameters in case of flow based capacity allocation.	B11
Critical network element market impact publication	A document providing the critical network elements and its impacts on the market to be published.	B12
Weather document	An estimation or prediction in advance of the weather by analysis of meteorological data and the results of what actually happened with the weather.	B13
Energy prognosis document	A document to provide the prognosis of energy production/load for a given period.	B14
Network constraint document	A document providing the network constraint situations used for the load flow studies. A network constraint situation includes contingencies, monitored elements and remedial actions.	B15
Aggregated netted external market schedule document	A document used to report aggregated netted external market schedules for a given border.	B16
Aggregated netted		

external TSO schedule document	A document used to report aggregated netted external TSO schedules for a given border.	B17
Reporting status market document	A document used to report the status of aggregated netted external market schedules, aggregated netted external TSO schedules and compensation program schedules on a given border.	B18
Reporting information market document	A document used to report the information concerning aggregated netted external schedules, aggregated netted external market schedules, aggregated netted external TSO schedules, compensation program schedules, netted area position schedules and netted area AC position schedules to an interested party.	B19
Status request for a reporting information market document	A document requesting the provision of a reporting information document.	B20
Reserve need document	Used by a TSO to send its reserve needs.	B21
Generation and load shift keys document	A document providing the values of the generation and load shift keys to be used on network model.	B22
Offers to be activated	A document containing the outcome of the process, with the list of offers that are to be activated by the TSO concerned and the results for its reserve needs.	B23
Clearing price	A document containing the outcome of the process, with the clearing prices for a period.	B24
Security analysis report	A document providing a report on a performed security analysis.	B25
Aggregated netted external schedule document	A document used to report aggregated netted external schedules for a given border.	B26
External TSO schedule	A document used to report external TSO schedules for a given border or interconnector.	B27
Move of scheduled production	A document indication a move of scheduled production.	B28
PS&LC results document	A document providing Pole Splitting & Loss Calculation results.	B29
Notification data market document	A document used to notify data to any information receiver.	B30
Additional Constraint document	A document describing additional constraints for a capacity calculation process.	B31
Operational state document	A document used for exchanging operational states for grid assets.	B32
Published offered capacity	A document providing the most recent values of offered capacity.	B33
Market result document	Published prices and volumes	B34
Area Configuration document	A document containing the definition of areas.	B35
Area Composition document	A document containing the relations between areas, i.e which Metering Grid Areas a Bidding Zone composed of.	B36
Connected Areas document	A document containing which other areas an area is connected to i.e. which Metering Grid Areas a Metering Grid Area is connected to.	B37
Settlement document	A document providing settlement information.	B38
Imbalance prognosis document	A document to provide the prognosis of energy imbalances for a given period.	B39
Complete set of bids	Submission of complete set of bids. If there are existing bids, they should be replaced.	B40
Merged MOL notice	A notice providing information on the MOL merging process.	B41
K-factor document	A document providing K-factor values.	B42
Settlement coordination document	A document providing settlement information for coordination between different parties.	B43
Financial settlement document	A document providing financial settlement information.	B44
Bid availability document	A document providing the detailed reasons for changing the availability or volume of a bid.	B45
Resource capacity unit document	A document providing information about resource capacity units.	B46
Other market information	A document providing other market information.	B47
Permission document	A document that contains detailed information about permissions.	B48
Requested capacity	A document providing requested capacity (e.g. ATC, CZCL) value.	B49
Voltage control activation document	A document providing the number of generating units to activate for voltage control, by means of their operation in synchronous condenser mode.	B50
Transmission inventory document	A document providing a transmission inventory.	B51
Transmission inventory	A document providing statistics on the transmission grid.	B52

Compounds

ESMP_DateTimeInterval

This datatype enables to express the start date and time, and the end date and time of a time interval with a specific pattern. This pattern is the YYYY-MM-DDThh:mmZ.

!# shows all attributes of ESMP_DateTimeInterval.

Table !# - Attributes of ESMPDataTypes::ESMP_DateTimeInterval.

Attribute name	mult.	Attribute type	Description
start (Compound)	1..1	YMDHM_DateTime	The start date and time of the interval with a minute resolution.
end (Compound)	1..1	YMDHM_DateTime	The end date and time of the interval with a minute resolution.

Primitives

String

A string consisting of a sequence of 8 bit characters. The character encoding is UTF-8. The string length is unspecified and unlimited.

DateTime

Date and time as "YYYY-MM-DDThh:mm:ss.sssZ", which conforms with ISO 8601 UTC time zone.
Note: the time within ESMP is expressed in UTC.

UUID

A UUID as defined by IEC/ISO 9348-8.

Integer

An integer number. The range is unspecified and not limited.

Decimal

Decimal is the base-10 notational system for representing real numbers.

Duration

Duration as "PnYnMnDTnHnMnS" which conforms to ISO 8601, where nY expresses a number of years, nM a number of months, nD a number of days. The letter T separates the date expression from the time expression and, after it, nH identifies a number of hours, nM a number of minutes and nS a number of seconds. The number of seconds could be expressed as a decimal number, but all other numbers are integers.